



Master in Computational Colour and Spectral Imaging (COSI)



Cross-Illumination Albedo Invariance: Lightness-Stable Intrinsic Decomposition from Real Multi-Illumination Photographs

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Abstract

Intrinsic image decomposition (IID) tries to separate a photograph into two layers: the *albedo*, which describes the reflectance of each surface, and the *shading*, which describes how light falls on those surfaces. A single photograph cannot tell a dark surface from a dimly lit one, or a warm surface from a white surface under warm light, so part of the lighting leaks into the recovered albedo. The same wall then comes out lighter or darker, and sometimes tinted, depending on how the room was lit, although the paint has not changed.

We train against this leakage with *Cross-Illumination Albedo Invariance* (CIAI). Two photographs of the same scene under different illumination pass through the same network; an invariance loss requires their predicted albedos to agree, and an explanation loss requires the ratio of the two predicted shadings to match the luminance ratio of the two images. The idea follows Siamese representation learning and multi-view consistency in 3D reconstruction, with positive pairs only. No dataset offers real photographs under several illuminations together with dense measured albedo, so we use the Multi-Illumination Dataset (MID), in which every scene is photographed 25 times with a flash bounced in a different direction, and the pseudo-ground-truth albedo that MIDIntrinsics derives for all 25 frames. CIAI is added as a second training stage to a model with a frozen DINOv2-Large encoder, a DPT decoder and three-channel shading; inference remains a single image and a single forward pass.

In matched ablations CIAI reduces the variation of a material’s recovered lightness across illuminations (C_{mat}) by 28% and 39%, and on synthetic ARAP renderings with their original coloured lighting it also lowers the dense albedo error by 7–14%. Our model is more lightness-stable than CD-IID, both Marigold variants and Ordinal Shading, level with CRefNet and behind RGB→X, while training 18.5M parameters. However, CIAI does not make the albedo stable in *colour*. The explanation loss is defined on luminance, and MID’s flash is white: between two frames the light changes mainly in direction and intensity, and its chromaticity moves by a median of only 0.030. Under CIAI the scale-normalised hue drift improves by only 4–6%, and our model’s hue drift still grows with the illuminant colour as much as that of a grey-shading method. CD-IID, which predicts the shading colour in a dedicated stage, leads both hue drift and chroma accuracy. We therefore propose, as the next stage, a shading split into a luminance and a chroma factor, a chroma explanation loss on the chroma factor, and training pairs whose illuminant colour genuinely varies.

Every stability result is reported next to accuracy against measured albedo (MAW) and against the MID pseudo-ground truth, because a grey, constant albedo is perfectly stable.

Note on this revised edition.. This edition corrects the version submitted in July 2026. That version described MID as a source of coloured-illuminant pairs and attributed the pooled chroma score to the literature; both were wrong. Its headline model came from a refinement study that changed several regularisers and the data mix at once; that study is removed, and the model reported throughout is now the complete CIAI configuration of the matched ablation. The follow-up colour studies are also removed, and the colour limitation is reported as a limitation with

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This project builds on the work of the intrinsic image decomposition community. I am grateful to the researchers who made their publications, datasets, and implementations publicly available. The code and supporting material for this thesis are available at <https://github.com/tmkhang1999/CIAI>.

Finally, this thesis marks the completion of my Master's studies in the COSI Erasmus Mundus programme. I thank the COSI board for its continued support and for enabling me to carry out this research in collaboration with 3DUniversum.

Glossary

Albedo	The intrinsic material colour of a surface, independent of the illumination. A white wall has high albedo in all three colour channels; a black wall has low albedo. In the IID model $\mathbf{I} = \mathbf{A} \cdot S$, the albedo is the term $\mathbf{A}$.
Chromaticity	The colour of a pixel or region expressed independently of its overall brightness. For an RGB pixel (r, g, b) , the chromaticity is typically the normalised vector $(r, g, b)/(r + g + b)$. A constancy metric based on chromaticity measures whether the hue of the predicted albedo shifts when the illuminant changes.
Colour cast	A global or regional shift in the hue of an image caused by a coloured light source. A warm orange lamp produces an orange cast; a fluorescent tube produces a green cast. When a cast leaks into the predicted albedo it means the model has not separated the illuminant colour from the material colour.
Illuminant	The light source or combination of light sources illuminating a scene. An illuminant is characterised by its spectral power distribution, which determines both its brightness and its colour (e.g. warm, cool, neutral).
Intrinsic Image Decomposition (IID)	The task of decomposing a photograph into an albedo layer (material colour) and a shading layer (lighting effect). See Chapter 3 for the formal definition.
Diffuse shading	The per-pixel quantity that encodes how much light falls on each surface point, combining the effect of surface normal orientation, light distance and direction, and occlusion (shadows). In this work the shading is a three-channel (colored) quantity $\mathbf{S}_d$, so it can represent a coloured illuminant; see Chapter 3.

Non-diffuse residual	The remaining image content not explained by the albedo-times-diffuse-shading product: specular highlights, visible light sources, and clipped regions. Computed analytically as $\mathbf{R} = (\mathbf{I} - \mathbf{A} \odot \mathbf{S}_d)_+$, with no dedicated network head.
White-balance (WB)	An image processing step that neutralises the colour of the illuminant, making images appear as if they were lit by a pure white light. White balance is applied by cameras automatically or in post-processing. In this work, per-frame WB is <i>disabled</i> for the MID training pairs so that the small illuminant colour difference between two flash directions is kept; it does not change intensity, so the brightness change between frames is present either way.
Scale-and-shift invariance (SSI)	A property of a metric or loss that makes it insensitive to global multiplicative and additive offsets. An SSI shading loss aligns the predicted shading to the ground truth using the best global scale and bias before computing the error, so it measures the quality of the relative shading pattern rather than its absolute level.

Acronyms

IID	Intrinsic Image Decomposition
CIAI	Cross-Illumination Albedo Invariance
MID	Multi-Illumination Dataset
ARAP	As Real As Possible (evaluation dataset)
MAW	Measured Albedo in the Wild
IIW	Intrinsic Images in the Wild
ViT	Vision Transformer
DPT	Dense Prediction Transformer
SSI	Scale-and-Shift Invariant
LMSE	Local Mean Square Error
si-RMSE	Scale-Invariant Root Mean Square Error
WHDR	Weighted Human Disagreement Rate
RCI	Reflectance and Chromaticity Inconsistency
RMSE	Root Mean Square Error
L1	Absolute Error (mean absolute deviation)
MSG	Multi-Scale Gradient
DSSIM	Structural Dissimilarity
WB	White Balance

Notations

$\mathbf{I} \in [0, 1]^{H \times W \times 3}$	Observed RGB image
$\mathbf{A} \in [0, 1]^{H \times W \times 3}$	Albedo (intrinsic material colour)
$\mathbf{S}_d \in \mathbb{R}_{>0}^{H \times W \times 3}$	Colored diffuse shading (three-channel)
$\mathbf{R} \in \mathbb{R}_{\geq 0}^{H \times W \times 3}$	Non-diffuse residual; $\mathbf{R} = (\mathbf{I} - \mathbf{A} \odot \mathbf{S}_d)_+$
$\boldsymbol{\pi} \in (0, 1)^{H \times W \times 3}$	Network shading output before inversion; $\mathbf{S}_d = (1 - \boldsymbol{\pi})/\boldsymbol{\pi}$
$(\mathbf{I}_1, \mathbf{I}_2)$	A cross-illumination pair: same scene photographed under two different illuminants
Ω_v	Set of valid paired pixels (excludes clipped/near-black pixels in either frame of a pair)
Ω_m, Ω_d	Set of pixels with valid ground-truth labels; set of pixels with diffuse-shading ground truth
$\nabla \mathbf{A}$	Spatial gradient of the albedo
$L(\cdot)$	Rec. 601 luminance of an RGB quantity
$\mathcal{L}_a$	Albedo ℓ_1 supervision loss
$\mathcal{L}_{\text{dssim}}$	Albedo structural-dissimilarity loss
$\mathcal{L}_{\text{msg-a}}, \mathcal{L}_{\text{msg-s}}, \mathcal{L}_{\text{msg-r}}$	Multi-scale gradient losses (albedo, shading, reconstruction)
$\mathcal{L}_{\text{chroma}}$	Albedo chroma-direction loss (hue/saturation, brightness-independent)
$\mathcal{L}_s$	Scale-and-shift invariant (SSI) shading loss
$\mathcal{L}_{\text{recon}}$	Diffuse reconstruction loss; $\ \mathbf{A} \odot \mathbf{S}_d - \mathbf{A}^* \odot \mathbf{S}_d^*\ _1$
$\mathcal{L}_{\text{res}}$	Residual closure and sparsity loss
$\mathcal{L}_{\text{mat}}$	DINOv2-token material-consistency loss
$\mathcal{L}_{\text{inv}}$	Cross-illumination invariance loss (CIAI, albedo consistency)
$\mathcal{L}_{\text{expl}}$	Shading explanation loss (CIAI, illuminant ratio)
$\mathcal{L}_{\text{sign}}$	Shading sign loss (positivity prior)
$\lambda_\star$	Weight of the corresponding loss term $\mathcal{L}_\star$
C_{mat}	MID chromaticity constancy metric (lower is better)
C_{arap}	ARAP constancy metric (lower is better)
Cast_{RMS}	RMS colour cast residual (lower is better)
WHDR	Weighted Human Disagreement Rate on IIW (lower is better)
ΔE	CIELAB colour difference on MAW (lower is better)

Contents

1	Introduction	1
1.1	Lighting Leaks into the Albedo	2
1.2	Our Approach: Cross-Illumination Albedo Invariance	3
1.3	Contributions	4
2	Related Work	7
2.1	Classical Intrinsic Image Decomposition	7
2.2	Learning-Based Intrinsic Image Decomposition	9
2.3	Multi-Illuminant Scenes and Colour Constancy	10
2.4	Evaluation Benchmarks	12
2.4.1	IIW	12
2.4.2	MID: Multi-Illuminant Dataset	13
2.4.3	ARAP	13
2.4.4	MAW: Measured Albedo in the Wild	14
3	Method	15
3.1	Physical Model	15
3.1.1	The Standard IID Equation	15
3.1.2	Why Achromatic Shading Fails with Coloured Light	15
3.1.3	The Shading Parameterisation	17
3.2	Network Architecture	17
3.2.1	DINOv2 Encoder	18
3.2.2	DPT Decoder	19
3.2.3	Output Heads	19
3.3	Cross-Illumination Albedo Invariance (CIAI)	20
3.3.1	The Core Idea	20
3.3.2	Invariance Loss	22
3.3.3	Explanation Loss	22
3.3.4	Raw Pairs and the Role of White Balance	23

CONTENTS

4	Implementation	25
4.1	Loss Terms	25
4.1.1	Supervised Anchor Losses	25
4.1.2	CIAI Losses	28
4.1.3	Physics and Regularisation Losses	29
4.2	Training Data	29
4.2.1	Hypersim	29
4.2.2	InteriorVerse	30
4.2.3	MID: Real Multi-Illuminant Pairs	30
4.3	Optimisation Schedule	31
5	Experiments	35
5.1	Experimental Setup	36
5.2	MID: Multi-Illuminant Constancy	39
5.3	IIW: Perceptual Reflectance Ordering	49
5.4	MAW: Measured Real-World Albedo	51
5.5	ARAP: Constancy and Albedo Accuracy	53
5.6	Ablation Study	60
5.7	Colour Stability: A Limitation	63
5.7.1	How much colour does MID’s illumination vary?	63
5.7.2	Does our hue drift follow the illuminant colour?	64
6	In Practice	67
6.1	Factor-Space Editing	68
7	Conclusion	71
7.1	Summary	71
7.2	Limitations	72
7.3	Future Work	72
	Bibliography	75
	List of Figures	79
	List of Tables	81

1 | Introduction

A photograph does not record what things look like. It records how light from those things reached the camera at one specific moment.

after Edwin Land

When we look at a red apple under a warm yellow lamp, we still see a red apple. Our visual system separates the colour of the apple’s skin from the colour of the light hitting it. This ability, called *colour constancy*, is so automatic in humans that we rarely think about it. For computers, it is a hard problem. A camera simply records the product of surface colour and illumination, and the two are mixed together in every pixel. Separating them is the goal of *Intrinsic Image Decomposition* (IID).

IID has a long history in computer vision and colour science, dating back to the Retinex theory of Land and McCann (1971). The core idea is that a photograph $\mathbf{I}$ can be modelled as the product of two layers: an *albedo* $\mathbf{A}$ and a *shading* S .

$$\mathbf{I}(p) = \mathbf{A}(p) \cdot S(p), \quad \forall p \in \Omega \tag{1.1}$$

Here p denotes a pixel, Ω is the set of all pixels, the albedo encodes the true material colour at each pixel, and the shading encodes how bright the surface is due to lighting and geometry. Throughout this report, “material colour” means this diffuse albedo in linear RGB, not the spectral reflectance of the surface. If we can recover $\mathbf{A}$ reliably, we obtain a lighting-independent description of the scene. This separation has practical consequences. An interior-design tool can recolour a wall without painting over its shadows; an augmented-reality system can place a virtual chair under the room’s existing light; a photograph can be re-lit without changing

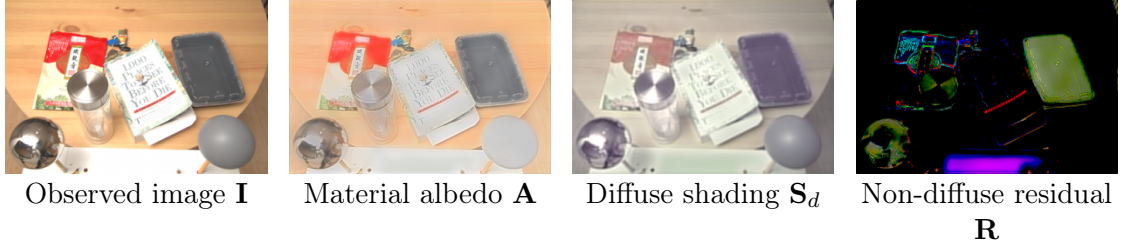


Figure 1.1: *Visual meaning of the decomposition used in this thesis. The photograph is explained by material colour multiplied by the illumination arriving at each surface, plus a residual for effects such as highlights and visible light sources. These are illustrative predictions, not separate photographs.*

the apparent material; and a catalogue or inspection system can compare product colours captured under different lamps. All four applications fail when the lighting, its brightness pattern or its colour, remains baked into the estimated material.

1.1 Lighting Leaks into the Albedo

Eq. (1.1) has twice as many unknowns as observations, so every IID method relies on priors to choose one factorisation. When those priors fail, part of the lighting is assigned to the albedo. Two forms of this leakage matter here.

The first is *lightness* leakage. A shadow, a soft gradient from a nearby lamp or a bright patch where light falls directly can be read as a darker or lighter material. The same wall then receives a different albedo depending on where the light comes from, even if the light is white.

The second is *colour* leakage. Most IID methods, classical and modern, assume a grey illuminant, so that the shading S in Eq. (1.1) is one value per pixel and all colour in the image is attributed to the albedo. Real indoor light is rarely grey: incandescent bulbs are warm, LEDs can be cool, and light bounced off a coloured wall takes on its colour. With a scalar shading the illuminant hue has nowhere to go but the albedo, and even with a three-channel shading a single image does not say how much of a hue belongs to the surface and how much to the light.

Both are instances of the same ambiguity, and both can in principle be broken by a second observation of the same scene under different light: the surfaces are identical in the two images, so whatever differs between them is lighting. Figure 1.2 shows the setting used in this work: one real scene photographed under several flash directions, and the albedo our model predicts from each photograph on its own.

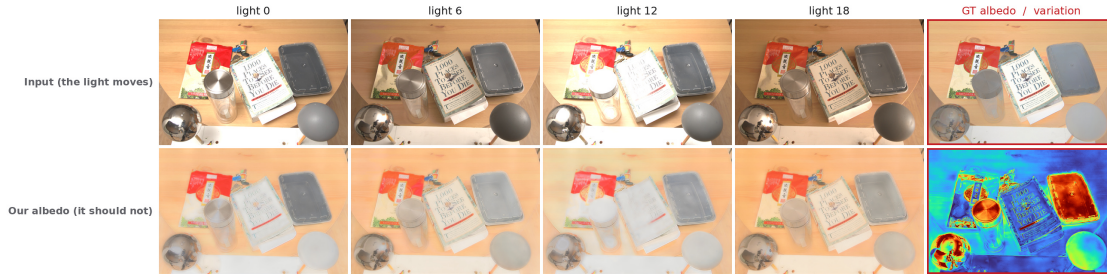


Figure 1.2: Albedo predictions on one MID scene under four flash directions. The upper row shows the four inputs and the dataset pseudo-ground-truth albedo; the lower row shows the four predictions and a per-pixel coefficient-of-variation map of predicted lightness (red marks regions that change with the light). The map stays cool over the table and most matte surfaces, and turns red on the plastic tray, the glass jar and the mirror sphere.

1.2 Our Approach: Cross-Illumination Albedo Invariance

We turn the second observation into a training signal and call it *Cross-Illumination Albedo Invariance* (CIAI). Two photographs of one scene under different illumination are passed through the same network. An invariance loss requires the two predicted albedos to agree, and an explanation loss requires the ratio of the two predicted shadings to match the luminance ratio of the two images, so that the model cannot satisfy the first loss with a flat albedo. The pairing is needed only during training; at inference the model takes one image.

The idea is borrowed from two places. Siamese and contrastive representation learning train a network so that two views of the same object map to the same representation (Chen and He, 2021); multi-view 3D reconstruction asks one surface, observed several times, to receive one set of properties. CIAI keeps only positive pairs, with no negatives, and replaces the learned embedding by a dense, physically defined target: the albedo map. Training IID on images of one scene under varying light is not new in itself; time-lapse and illumination-varying sequences have been used for self-supervised decomposition (Li and Snavely, 2018b; Lettry et al., 2018; Ma et al., 2018; Kinoshita and Kiya, 2021). What we study is the effect of this constraint as a fine-tuning stage on top of supervised synthetic training in a modern feed-forward model, using real raw photographs, and what such pairs can and cannot teach.

Applying CIAI requires real photographs of one scene under several illuminations, and no dataset combines them with dense measured albedo. We use the Multi-Illumination Dataset (MID) (Murmann et al., 2019), in which each scene

is photographed 25 times from a fixed camera with a flash bounced in a different direction, together with the pseudo-ground-truth albedo that MIDIntrinsics derives for all 25 frames of a scene (Careaga and Aksoy, 2023). The flash is white. Between two MID frames the light therefore changes mainly in direction and intensity, and its colour changes only through light bounced off coloured surfaces; we measure a median chromaticity gap of 0.030 between frames. MID is a strong signal for lightness and a weak one for colour, and this property turns out to decide what CIAI can learn.

We implement CIAI inside a model built on a frozen *DINOv2-Large* vision encoder (Oquab et al., 2024) and a Dense Prediction Transformer (DPT) decoder (Ranftl et al., 2021) with a three-channel shading output. The model is first trained with supervised losses on Hypersim (Roberts et al., 2021) and InteriorVerse; CIAI is added in a second stage on MID pairs.

1.3 Contributions

The contributions of this work are:

- **A paired training strategy for lightness stability.** An albedo-invariance loss and a luminance explanation loss on raw MID pairs, used as a fine-tuning stage. Matched ablations show that it lowers the variation of recovered material lightness across illuminations by 28% and 39%, and lowers the dense albedo error on raw, coloured-lit ARAP renderings by 7–14%, a benchmark it never trained on. The model trains 18.5M parameters on top of a frozen encoder.
- **A matched ablation.** Turning the cross-illumination losses and the colour path on and off from one shared checkpoint isolates the effect of CIAI: a large gain in lightness stability, a small gain in hue stability, and a lower albedo error on raw ARAP renderings. The colour limit follows from the luminance-only explanation loss and MID’s white flash, and we measure it directly.
- **An evaluation that reports stability next to accuracy.** Material-level lightness and hue stability on MID, always read beside accuracy against measured albedo (MAW) and against the MID pseudo-ground truth, because a grey, constant albedo is perfectly stable.
- **A plan for the colour axis.** A three-factor model that separates shading luminance from shading chroma, a chroma explanation loss on that factor,

rendered and real training pairs whose illuminant colour actually varies, and a cross-architecture test of the training strategy.

The project code, training configurations, and evaluation scripts are publicly available at <https://github.com/tmkhang1999/CIAI>.

The rest of this report is organised as follows. Chapter 2 reviews related work on IID, paired-illumination training, multi-illuminant scenes, and the evaluation benchmarks. Chapter 3 describes the model architecture and the CIAI training strategy. Chapter 4 covers the loss terms, training data, and optimisation schedule in detail. Chapter 5 presents the benchmark results, the ablation, and the colour limitation. Chapter 6 demonstrates qualitative factor edits built directly on the decomposition. Chapter 7 summarises the findings, discusses limitations, and sets out the next stage of the work.

2 | Related Work

The history of science is the history of struggles against the default assumption.

Anonymous

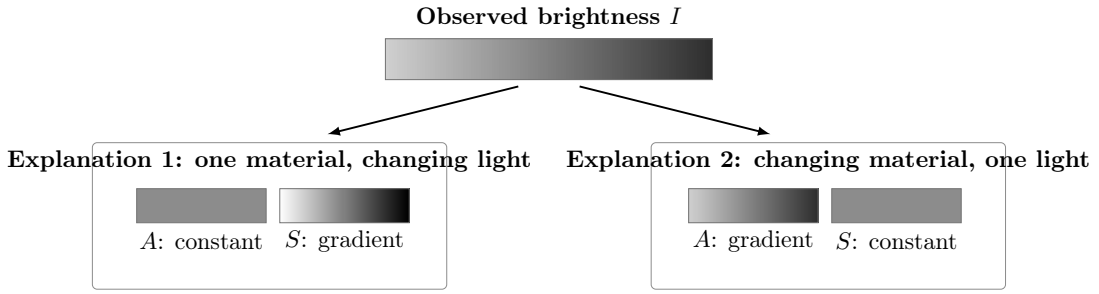
This chapter reviews prior work that is relevant to the approach taken in this thesis. We cover classical and learning-based IID methods in Sections 2.1 and 2.2, work on multi-illuminant scenes and colour constancy in Sect. 2.3, and the evaluation benchmarks we use in Sect. 2.4.

2.1 Classical Intrinsic Image Decomposition

The problem of separating illumination from reflectance is rooted in Retinex theory (Land and McCann, 1971). Its starting observation is that perceived lightness depends strongly on spatial comparisons, not only on the number recorded at one pixel. If neighbouring image intensities are $I_1 = A_1 S_1$ and $I_2 = A_2 S_2$, their ratio is

$$\frac{I_1}{I_2} = \frac{A_1}{A_2} \frac{S_1}{S_2}. \quad (2.1)$$

When illumination varies slowly over that small neighbourhood, $S_1/S_2 \approx 1$, so the observed ratio is evidence for a reflectance change. For example, a painted black stripe on a white wall should remain a material boundary as the room becomes brighter. A soft illumination gradient should not. Retinex-style methods classify or threshold such local changes and propagate the surviving ratios to estimate a reflectance field.



Both satisfy $I = AS$; a single image cannot identify which explanation is correct.

Figure 2.1: *The intrinsic-image ambiguity in one dimension. The same observed brightness gradient can be explained by constant reflectance under changing illumination (left), or by changing reflectance under constant illumination (right). Retinex and learned IID methods resolve this underdetermined factorisation by adding spatial, semantic, and data-derived priors.*

This is foundational because it turns an impossible per-pixel factorisation into a problem that can use spatial context. It is also only a prior. A hard cast shadow can change abruptly and resemble paint; textured material can vary rapidly; specular highlights violate the diffuse product model. The common shorthand “illumination is low-frequency and reflectance is high-frequency” is therefore a useful intuition, not a physical rule. Modern learned methods replace many hand-designed thresholds with data, but they still rely on the same distinction between material changes and lighting changes.

Figure 2.1 shows why such assumptions are necessary. Even a simple brightness gradient has multiple exact decompositions: the image alone does not reveal whether the change came from the surface or the light.

Grosse et al. (2009) introduced the MIT Intrinsic Images dataset, the first public benchmark for IID, and used it to show that simple Retinex baselines outperformed more sophisticated methods of the time on real photographs. A consistent finding from this benchmark is that the problem is extremely ill-posed: many different (albedo, shading) pairs explain the same image. Without strong priors, models tend to find trivial solutions.

The most influential approach to adding structure to the ill-posed problem is Shape, Illumination, and Reflectance from Shading (SIRFS) by Barron and Malik (2015). SIRFS frames IID as an energy minimisation over albedo, shading, and surface normals jointly. It encodes strong priors: albedos should be piecewise-constant, shadings should be smooth, and surface normals should form a plausible scene geometry. SIRFS significantly outperformed previous methods and remained a strong baseline for several years. However, its hand-designed priors struggle on complex real-world scenes with cluttered backgrounds, unusual lighting, or

materials that violate the piecewise-constant albedo assumption.

2.2 Learning-Based Intrinsic Image Decomposition

The availability of large synthetic datasets shifted the field toward learned models. Li and Snavely (2018a) proposed CGIntrinsics, a set of physically-based re-renderings of synthetic indoor scenes that provides paired (image, albedo, shading) training data at scale. They showed that a CNN trained on this data, combined with ordinal supervision from the human judgements of IIW (Bell et al., 2014), generalises well to real photographs. CGIntrinsics established the pattern that has dominated the field since: train on photorealistic synthetic data with ground-truth labels, then evaluate on real images.

Bell et al. (2014) introduced the Intrinsic Images in the Wild (IIW) benchmark, which takes a different approach to ground truth. Rather than rendering or measuring true albedos, IIW collects human judgements about relative albedo comparisons: for any two surface patches in a photograph, human annotators say which one is lighter. This produces a set of relative constraints that any valid albedo prediction must satisfy. The evaluation metric, the Weighted Human Disagreement Rate (WHDR), measures how often the model’s albedo ordering disagrees with the majority of human annotators. IIW became the standard benchmark for IID for many years and is still widely used. Its limitation is that it captures perceptual shading quality (whether shadows look right) more than it captures the accuracy of the recovered material colour.

A line of work by Careaga and Aksoy is the closest to ours in its treatment of illumination colour. Ordinal Shading (Careaga and Aksoy, 2023) decomposes real photographs by first estimating shading up to an ordinal (ordering) constraint, which is far more robust than absolute regression, and then refining the albedo; its shading is still greyscale. Its successor, Colorful Diffuse IID (CD-IID) (Careaga and Aksoy, 2024), is the first method to explicitly predict a *colored* diffuse shading in the wild, through a multi-stage cascade (grey decomposition, then a low-resolution chromaticity network, then albedo and diffuse-shading refinement) and an analytic non-diffuse residual. CD-IID is the direct point of comparison for this work. It trains each stage on single images, including MIDIntrinsics, and uses no loss that ties the albedos of two illuminations of one scene together. Its colour separation instead comes from the architecture: a dedicated stage that predicts the chromaticity of the shading. Our measurements in Chapter 5 suggest that this structural choice, rather than any amount of paired supervision on MID, is what makes its albedo colour-stable, and the model we plan next adopts the same split.

In parallel, diffusion-based models have become strong IID baselines. Marigold-IID (Ke et al., 2025) fine-tunes a Stable-Diffusion backbone for IID and ships two variants: an *appearance* model trained on InteriorVerse that targets material properties (albedo, roughness, metallic), and a *lighting* model trained on Hypersim (Roberts et al., 2021) that targets albedo, diffuse shading, and a non-diffuse residual. Both produce visually sharp decompositions and excellent zero-shot WHDR on IIW. We use Marigold-IID as our primary experimental baseline because checkpoints are public, its lighting variant shares our model’s training distribution (Hypersim), and it represents the current generative state of the art. Very recent work reaches for external knowledge instead: ReasonX (Dirik et al., 2025) guides the decomposition with a multimodal language model and reports strong IIW WHDR improvements; no public checkpoint was available at the time of writing, so we cite its published numbers only where protocols match.

Comparison scope.. The methods above are not all compared in the same way. We run models locally only when an official checkpoint and compatible output are available: Marigold-IID, CRefNet, Ordinal Shading, CD-IID and RGB→X (Zeng et al., 2024). We cite ReasonX and older methods from their papers only when the dataset split and metric match our protocol. This distinction matters because a published number obtained with different training data, resizing, or test scenes is context, not a controlled head-to-head result. Chapter 5 marks locally reproduced and literature-cited rows separately and explains why each model is included.

What these methods share is that each image is decomposed, and supervised, on its own. Nothing in a single-image objective tells the model that two photographs of the same room should yield the same albedo, so whatever lighting the priors fail to explain, in brightness or in colour, can remain in the albedo and differ from one photograph to the next. Our work adds that cross-image constraint during training, using real multi-illumination photographs, and measures what it fixes and what it does not.

2.3 Multi-Illuminant Scenes and Colour Constancy

The specific challenge of coloured illumination in IID has received relatively little attention compared to the broader IID problem. Most methods implicitly assume a single white or near-white illuminant.

The cost of that assumption is arithmetic rather than model-specific. If shading is scalar, it has no hue, so a coloured illuminant must be represented in the only

RGB factor: albedo. Chapter 3 derives this limitation and demonstrates it visually in Figure 3.1; this section focuses on how prior work has responded to it.

Work on multi-illuminant colour constancy in computational photography provides useful context. Beigpour et al. (2014) addressed the problem of estimating multiple coloured illuminants from a single image using a conditional random field formulation. They created the MIID dataset, which contains scenes photographed under combinations of coloured light sources with ground-truth illuminant annotations. This is a different task from IID (they estimate illuminant colours rather than separate albedo from shading), but the setup is closely related and provides inspiration for the multi-illuminant pair training we use in this work.

Training on one scene under several illuminations.. Using several images of a static scene under changing light is an old idea in IID: the reflectance is shared across the images while the illumination is not. Li and Snavely (2018b) learn a single-image decomposition from time-lapse sequences with losses that require a common reflectance across frames. Lettry et al. (2018) train a Siamese network on illumination-varying sequences, requiring the reflectances of two frames to agree and their cross-combined factors to reconstruct the images. Ma et al. (2018) train a two-stream network without ground-truth intrinsic images on pairs of the same scene under different illumination, and reduce it to a single stream at inference. These works use the paired constraint as the *main* supervision, mostly on time-lapse or synthetic sequences. We use it instead as a second training stage on top of supervised synthetic training, on real raw multi-illumination photographs, and measure separately what it does to lightness and to colour. The constraint is also close in spirit to Siamese representation learning without negatives (Chen and He, 2021), where two views of one object must map to one representation.

The closest prior work in its treatment of illuminant colour is the self-supervised decomposition network of Kinoshita and Kiya (2021). They observe the same gap we do: most IID methods do not enforce any consistency on the *reflectance* layer, and the usual white-illuminant model cannot express a coloured light in the first place. Their answer is to adopt a colour-illuminant model and to train with losses computed over several illumination conditions of the same content, so that the predicted reflectance is pushed to agree across them. The difference from our work lies in where those conditions come from. Their illumination variants are *simulated*, which makes the supervision cheap but ties it to the realism of the simulator. We instead take the constraint from real photographs of real rooms (Sect. 2.4.2). The cost of that choice, measured in Chapter 5, is that the real corpus available varies the direction and intensity of a white light far more than its colour.

This family of work also suggests the shape of a sensible metric. If a model is genuinely illuminant-invariant, the chromaticity it assigns to one physical material

must not move when the light changes. Our stability measures (C_{mat} and Cast_{rel} on MID, C_{arap} on ARAP) are built directly on that statement: they group the renderings or photographs of one scene by illumination and report how much the predicted albedo of each material drifts across the group (Sect. 5.2). They are our own diagnostics rather than a published protocol, which is why we report them alongside, not instead of, accuracy against measured albedo.

2.4 Evaluation Benchmarks

We evaluate our model on four benchmarks that together cover different aspects of IID quality.

Dataset, split, protocol, and benchmark.. These terms are kept distinct throughout the thesis. A *dataset* is a collection of images and annotations. A *split* identifies which samples are used for training, validation, or testing. An evaluation *protocol* specifies preprocessing, eligible samples, and a metric. A *benchmark* is the held-out data together with that protocol. Thus MID is both a source of training pairs and an evaluation dataset, but the CIAI training scenes and the 30-scene test split never overlap. Calling a collection a “database” adds no useful distinction here, so this thesis uses *dataset* consistently.

The four benchmarks answer different questions. MID tests whether a real surface stays stable across lights; MAW measures real material colour on sampled regions; ARAP provides dense synthetic albedo and repeated illumination; IIW tests whether predicted relative reflectance agrees with people. No single score is treated as a complete definition of IID quality.

2.4.1 IIW

Intrinsic Images in the Wild (Bell et al., 2014) is the most widely used benchmark in the IID literature. It contains 5,230 real indoor and outdoor photographs collected from the internet, annotated with human pairwise reflectance comparisons. The evaluation metric is WHDR: the fraction of annotator pairs (weighted by confidence) with which the model’s albedo ordering disagrees. Lower WHDR is better. IIW rewards agreement with perceived relative reflectance. Its human labels are appropriate for that perceptual question, but the ordinal lightness comparison does not measure RGB colour, absolute albedo, or dense texture. We evaluate on the standard test split of 1,046 images (every fifth image of the 5,230), and report WHDR as the mean over images rather than pooled over judgements.

2.4.2 MID: Multi-Illuminant Dataset

MID, the Multi-Illumination Dataset of Murmann et al. (2019), is the dataset most directly relevant to this work. It contains over 1,000 real indoor scenes, each photographed 25 times from a fixed viewpoint with a flash bounced in a different direction for every shot, and a grey and a chrome probe in every frame. The flash is white, so between two frames the light changes mainly in *direction and intensity*. Its colour changes only because the bounced light picks up the colour of the wall or ceiling it reflects from: measured on the grey probes, the median chromaticity distance between the illuminants of two frames of a scene is 0.030, and 86% of frame pairs lie below 0.08 (Sect. 5.7). The key property is that the albedo of each surface does not change across the 25 frames; only the light does. MIDIntrinsics (Careaga and Aksoy, 2023) adds one dense pseudo-ground-truth albedo per scene, derived from the white-balanced capture stack and shared by all 25 frames. This makes it possible to measure how much a model’s predicted albedo of each material changes across frames (C_{mat} for lightness, Cast_{rel} for hue), and how far it lies from the pseudo-ground truth. A perfectly illumination-invariant model would have zero variation. We hold out 30 scenes as a test split and report constancy on those; the training split is used for the CIAI pairs (Chapter 4).

2.4.3 ARAP

ARAP (“As Real As Possible”) is a synthetic high-dynamic-range dataset of around 50 rendered indoor and outdoor scenes (Bonneel et al., 2017). A subset of scenes ships with multiple renderings under different lighting conditions (direction, intensity, and in some scenes light colour); we evaluate constancy on the 37 multi-light scene groups (23 indoor), and additionally report the coloured-illuminant subset in which the light *colour* varies across renderings, which is the axis this thesis targets. ARAP provides ground-truth albedo images, which allows measuring not only stability across conditions (the C_{arap} metric) but also the accuracy of the predicted albedo against the ground truth. The ground truth is, however, stored in three different encodings across scenes, which our evaluation mask did not account for; the ARAP numbers in Chapter 5 are therefore provisional (Sect. 5.5). The constancy metrics group scene renderings by the illuminant condition and measure how consistent the predicted albedo is across groups. Our local study reports LMSE, SSIM, and a mean-normalised RMSE diagnostic. This implementation is not the least-squares scale-invariant RMSE used in published ARAP tables, so Chapter 5 keeps the local and published protocols separate. The two types of metric capture different things: a model can have good constancy (stable albedo across illuminants) while still being wrong about the absolute material colour, and

vice versa.

2.4.4 MAW: Measured Albedo in the Wild

MAW (Wu et al., 2023) is a benchmark of real indoor images paired with ground-truth albedo measurements obtained through a controlled physical measurement process. It provides a direct measure of albedo colour accuracy through two metrics: ΔE (the CIELAB colour difference between predicted and measured albedo) and intensity SI-MSE (scale-invariant mean square error on the luminance channel). ΔE captures colour error in a perceptually motivated space after the official scorer’s per-material scale alignment. MAW is particularly valuable because it measures real scenes where the illuminant colour is part of the challenge, making it complementary to the synthetic ARAP. We evaluate on 874 images from the MAW test split.

3 | Method

This chapter describes how our model works. We start with the physical model that underlies IID (Sect. 3.1), then describe the network architecture (Sect. 3.2), and finally explain the CIAI training strategy (Sect. 3.3).

3.1 Physical Model

3.1.1 The Standard IID Equation

Under the Lambertian reflectance model with a single distant light source, the intensity at a pixel p in an image is:

$$\mathbf{I}(p) = \mathbf{A}(p) \cdot S(p) \tag{3.1}$$

where $\mathbf{A}(p) \in [0, 1]^3$ is the three-channel albedo (material reflectance) and $S(p) \in \mathbb{R}_{>0}$ is a non-negative scalar shading value. Shading describes the illumination arriving at the surface point after geometry, visibility, and shadows have had their effect. Albedo describes the fraction and colour of that light that the material diffusely reflects. The camera observes the reflected result, not either factor in isolation.

Under the assumption of a grey (achromatic) illuminant, S is a scalar and the product in Eq. (3.1) is a simple element-wise multiplication: each colour channel of $\mathbf{I}$ is scaled by the same amount. This makes the colour of each pixel determined entirely by $\mathbf{A}$.

3.1.2 Why Achromatic Shading Fails with Coloured Light

When the illuminant has colour, the scaling is no longer the same across channels. A red lamp makes the R channel brighter relative to G and B . A correct model would encode this in the shading, but scalar shading cannot express different scalings

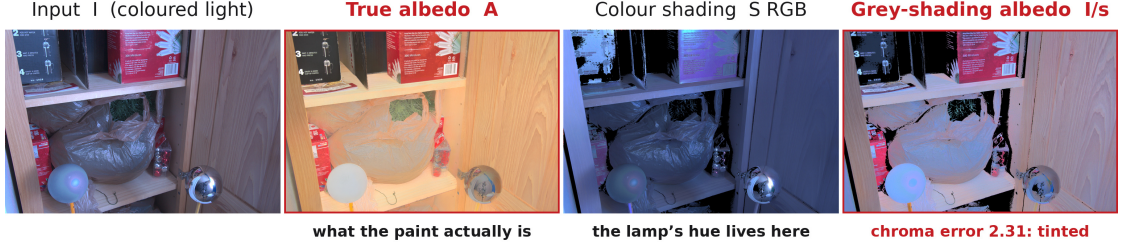


Figure 3.1: Why scalar shading must leak illuminant colour into albedo. From left: a warm-lit MID frame; pseudo-ground-truth albedo; the RGB shading implied by the data; and the albedo recovered with the luminance of that true shading. The favourable scalar construction still produces a visibly tinted albedo (mean chroma error 2.31, versus 0.22 when shading has three channels), because a scalar can alter brightness but not hue. The remaining RGB error comes from clipping and the display-domain conversion, not from an inability to represent channel-dependent illumination.

per channel. What actually happens in practice is that the model absorbs the channel imbalance into the albedo, because the albedo is the only RGB term in the equation. The predicted albedo therefore reflects the combined colour of the surface *and* the illuminant, rather than just the surface.

Figure 3.1 demonstrates that this is a limitation of the representation, not a particular network. Even when the scalar model is given the luminance of the true RGB shading, dividing the image by that scalar can change brightness but cannot remove the lamp’s hue.

To model coloured illumination correctly, the shading must therefore be a three-channel (RGB) quantity $\mathbf{S}_d(p) \in \mathbb{R}_{>0}^3$, so each colour channel can be scaled independently. The model additionally keeps an explicit non-diffuse term for the light that the Lambertian product cannot represent (specular highlights, visible light sources, clipped pixels), giving the full decomposition:

$$\mathbf{I}(p) = \mathbf{A}(p) \odot \mathbf{S}_d(p) + \mathbf{R}(p) \quad (3.2)$$

where $\odot$ is element-wise multiplication, $\mathbf{S}_d$ is the colored *diffuse* shading, and $\mathbf{R} \geq 0$ is the non-diffuse residual. The model predicts $\mathbf{A}$ and $\mathbf{S}_d$ with two network heads and derives the residual *analytically* as $\mathbf{R} = (\mathbf{I} - \mathbf{A} \odot \mathbf{S}_d)_+$, so the reconstruction $\mathbf{A} \odot \mathbf{S}_d + \mathbf{R}$ matches the input by construction wherever the diffuse product does not overshoot it.

A three-channel shading solves the representation problem but not the *ambiguity* problem: single-image supervision cannot say how much of a hue belongs to the albedo and how much to the shading colour, and the standard ground-truth datasets (including Hypersim) supervise shading only up to this ambiguity. Paired

observations can narrow this ambiguity. If the albedo is required to be identical across two images of a scene whose illuminant colours differ, any part of the illuminant colour that *differs* between the two frames can no longer sit in the albedo. Two limits follow directly, and both turn out to matter. A colour common to both frames (a constant tint) satisfies the constraint and is left unresolved; and the constraint can only remove colour differences that the training pairs actually contain. Sect. 3.3 defines the losses and Sect. 5.7 measures how much of this the real pairs deliver.

3.1.3 The Shading Parameterisation

A practical challenge in predicting shading is that it can take any positive value, which is an unbounded output range; worse, its distribution in real scenes is heavily skewed. Specular surfaces and bright light sources produce very large shading values, so linear shading concentrates almost all pixels in a narrow band near zero with a long tail (Figure 3.2): a regression loss on this representation is dominated by a few extreme pixels while the mid-range values that carry the shadow structure receive almost no gradient. Working in the logarithmic domain balances the distribution but still has no well-defined range for a network output. Following the inverse-shading parameterisation of Careaga and Aksoy (2023), which was introduced to solve exactly this conditioning problem, the network predicts a bounded quantity $\pi(p) = 1/(\mathbf{S}_d(p) + 1) \in (0, 1)$ per channel using a sigmoid activation, and the shading is recovered by inverting this mapping:

$$\mathbf{S}_d(p) = \frac{1 - \pi(p)}{\pi(p)} \quad (3.3)$$

The prediction is clamped to $\pi \in [5 \times 10^{-3}, 1 - 10^{-4}]$, which bounds the recovered shading to $\mathbf{S}_d \in [10^{-4}, 199]$, providing enough headroom to represent bright high-dynamic-range illumination while keeping the division numerically safe. The inversion is differentiable, so gradients flow through it during training. Losses that supervise the shading pattern are computed on π (bounded, well-conditioned), while physics losses that need the actual light intensity (reconstruction, the residual) use the linear $\mathbf{S}_d$.

3.2 Network Architecture

Our model is built on two components: a frozen DINOv2-Large visual encoder and a DPT-style decoder with separate output heads for albedo and shading.

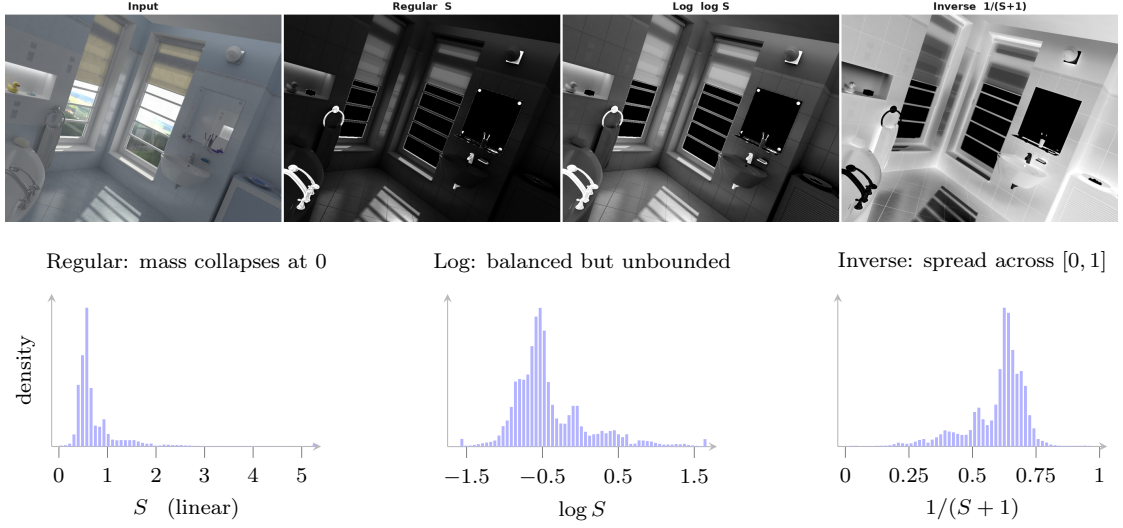


Figure 3.2: Why shading is regressed in the inverse domain. Shading is derived from *Hypersim* ground truth as $S = \mathbf{I}/\mathbf{A}$ and shown in three representations, with the distribution of its values beneath each. Linear shading is dominated by a small number of very bright pixels. It runs to 5.3 on this frame, so almost all of the mass collapses into the first bin and a regressor spends its capacity on outliers. Log shading balances the distribution but has no bounded range, so it cannot be produced by a saturating output nonlinearity. Inverse shading, $1/(S+1)$, is confined to $[0, 1]$ by construction and spreads across it, which makes it a well-conditioned target for a sigmoid head: this is the representation our shading branch predicts (Sect. 3.1), following Careaga and Aksoy (2023). The physics losses that need real light intensity operate on the linear $\mathbf{S}_d$ recovered from it.

3.2.1 DINOv2 Encoder

The encoder is DINOv2-Large (Oquab et al., 2024), a Vision Transformer (ViT) with a patch size of 14 pixels and approximately 304 million parameters. DINOv2 was trained with self-supervised learning on a large and diverse image corpus, and it produces patch-level feature representations that capture rich semantic and geometric information.

We use DINOv2 in a frozen state, meaning its weights are not updated during IID training. This is a deliberate design choice for two reasons. First, DINOv2’s representations are already very strong; fine-tuning them would require careful learning rate scheduling and risks destroying useful features learned during self-supervised pre-training. Second, keeping the encoder frozen reduces GPU memory during training, which allows us to use a larger batch size.

For an input image of size $H \times W$, DINOv2 divides it into $(H/14) \times (W/14)$ patches and produces a token embedding for each patch. We extract the intermediate features from four roughly evenly-spaced transformer blocks (blocks 5, 12, 18, and 24 of the 24-block ViT-L, the last of which is the final block). These multi-depth features capture information at different levels of abstraction: earlier blocks retain more local texture and colour, while later blocks encode semantic and structural information. One consequence of DINOv2’s heavily augmented self-supervised training is directly useful for IID: because its objective is invariant to strong photometric augmentation, its patch tokens are expected to be robust to low-level appearance changes including illumination, so a shadowed and a lit patch of the same material tend to map to similar tokens, which is a helpful prior for assigning the same albedo across a shadow boundary.

3.2.2 DPT Decoder

The Dense Prediction Transformer (DPT) decoder (Ranftl et al., 2021) was designed to reconstruct full-resolution feature maps from the patch tokens of a ViT. The key idea is to re-assemble the patch tokens from each layer into a spatial feature map, then fuse the maps from all layers through a series of convolutional feature fusion blocks. The fusion progressively upsamples the features to the input resolution, combining coarse semantic information from deep layers with fine spatial details from early layers.

We use a single DPT decoder that takes all four feature levels as input. Each level is first *reassembled*: the patch tokens are projected and reshaped into a spatial feature map at a level-specific resolution. The four maps are then merged by a chain of *fusion* blocks, each of which upsamples the coarser map, adds the next finer one, and refines the sum with residual convolutions, so semantic context from deep blocks and spatial precision from early blocks accumulate into one representation. Because the ViT operates on 14×14 patches, even the finest token map is far below pixel resolution; a lightweight convolutional *detail stem* therefore processes the raw input image in parallel and injects genuine high-frequency edge information that the token path cannot carry. The decoder output is a shared full-resolution feature map, which is fed to two separate output heads.

3.2.3 Output Heads

The albedo head takes the DPT feature map, concatenates a full-resolution *RGB skip* (the gamma-encoded input image), and passes the result through a small

convolutional block followed by a sigmoid activation:

$$\mathbf{A} = \sigma\left(\text{Conv}(\mathbf{F}_{\text{DPT}}, \mathbf{I}^{1/2.2})\right) \in [0, 1]^{H \times W \times 3} \quad (3.4)$$

The sigmoid bounds the albedo to $[0, 1]^3$, the correct range for a reflectance value. The RGB skip, together with the chroma-direction loss on albedo of Sect. 4.1, forms what we call the *colour path*. It is a design choice of this model and not part of CIAI: the frozen semantic tokens do not seem to carry all of the local, absolute colour needed for albedo, and without the skip the predicted albedo keeps only about half of the between-material chroma spread of the reference albedo (Sect. 5.6); we did not probe the encoder directly, so the reason is a hypothesis. The skip has a cost as well: it is a direct route by which the *illuminant* colour of the input can reach the albedo as a colour cast, which the invariance loss must then counteract. The colour path is one of the two ablation axes in Chapter 5.

The shading head applies its own convolutional block and sigmoid and outputs the three-channel inverse shading:

$$\boldsymbol{\pi} = \sigma\left(\text{Conv}(\mathbf{F}_{\text{DPT}})\right) \in (0, 1)^{H \times W \times 3} \quad (3.5)$$

The colored diffuse shading $\mathbf{S}_d$ is recovered from $\boldsymbol{\pi}$ using Eq. (3.3), and the non-diffuse residual follows analytically:

$$\mathbf{R} = (\mathbf{I} - \mathbf{A} \odot \mathbf{S}_d)_+ \quad (3.6)$$

There is no third head: the residual costs no parameters, and the reconstruction $\mathbf{A} \odot \mathbf{S}_d + \mathbf{R}$ is exact wherever the diffuse product does not overshoot the image.

Both heads share the same DPT feature map as input. This encourages the two predictions to be consistent with each other, since they are both derived from the same underlying scene representation.

3.3 Cross-Illumination Albedo Invariance (CIAI)

3.3.1 The Core Idea

CIAI is a training strategy, not a change to the network. It adds two loss terms that operate on *pairs* of images of the same scene photographed under different illumination. The design follows the Siamese pattern of representation learning (Chen and He, 2021): one network with shared weights processes both views, and only positive pairs are used. Unlike a learned embedding, the quantity the two views must share here is a dense physical map, the albedo.

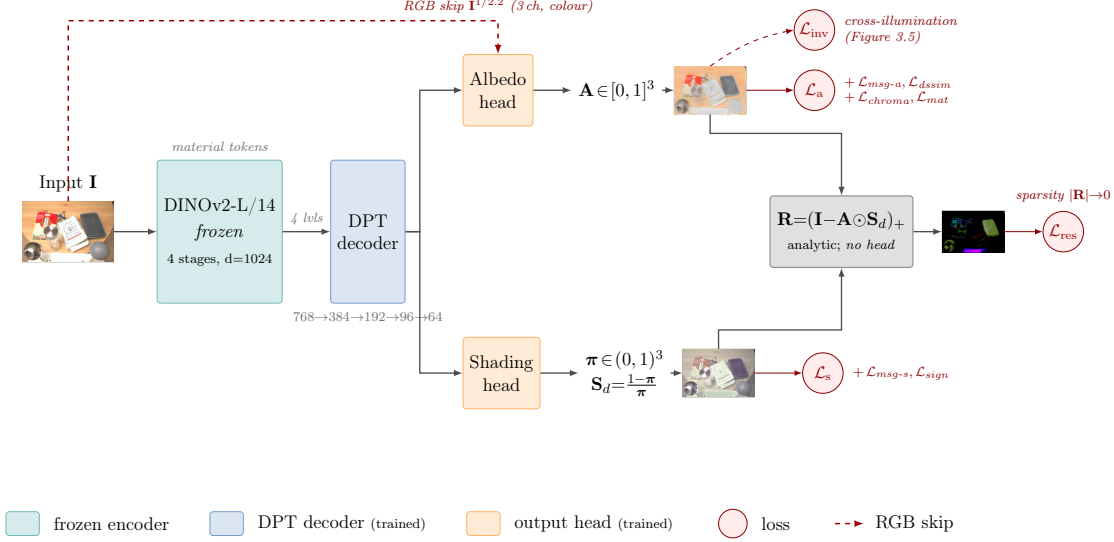


Figure 3.3: Model architecture and its supervision. A frozen DINOv2-Large encoder (teal) extracts patch features from four transformer stages; it is never fine-tuned, so only 18.5M parameters downstream are learned. A DPT decoder (blue) reassembles and fuses the four levels into a shared full-resolution feature map. Two output heads (amber) read it. The albedo head additionally takes a full-resolution RGB skip: the gamma-encoded input image, concatenated in directly. With the chroma loss on albedo it forms the colour path, which restores the material colour the frozen-encoder model otherwise washes out (Sect. 5.6); it is also a direct route for illuminant colour into $\mathbf{A}$, which is why the planned model replaces the raw input with the input divided by the predicted shading. The shading head has no such skip; it predicts the inverse shading π from the shared DPT features alone, from which $\mathbf{S}_d = (1 - \pi)/\pi$ follows. The non-diffuse residual $\mathbf{R} = (\mathbf{I} - \mathbf{A} \odot \mathbf{S}_d)_+$ is analytic; there is no third head. Red nodes are the training losses on each output: the albedo family ($\mathcal{L}_a$ with its multi-scale-gradient, structural, chroma and material-consistency terms), the shading family ($\mathcal{L}_s$ with its gradient and sign terms), and a sparsity term $\mathcal{L}_{res}$ that pushes the analytic residual toward zero. The cross-illumination invariance loss $\mathcal{L}_{inv}$ also constrains the albedo but acts across an illuminant pair at training time only (Figure 3.5). Thumbnails are this model’s own outputs on a MID test scene.

Let $(\mathbf{I}_1, \mathbf{I}_2)$ be such a pair, where $\mathbf{I}_1$ and $\mathbf{I}_2$ show the same scene under two different illuminations; on MID, two different flash directions. Let $\mathbf{A}_1 = \mathbf{A}(\mathbf{I}_1)$ and $\mathbf{A}_2 = \mathbf{A}(\mathbf{I}_2)$ be the albedos predicted by the model for each image, and let S_1 and S_2 be the corresponding shadings. Figure 3.5 at the end of this section shows the complete mechanism, once both losses below have been defined.

3.3.2 Invariance Loss

The albedo must be the same for both images, because the physical material has not changed. We encode this as a masked ℓ_1 loss:

$$\mathcal{L}_{\text{inv}} = \frac{1}{|\Omega_v|} \sum_{p \in \Omega_v} \|\mathbf{A}_1(p) - \mathbf{A}_2(p)\|_1 \quad (3.7)$$

where $\Omega_v \subset \Omega$ is the set of *valid paired pixels*. The validity mask matters in practice: MID frames are high-dynamic-range captures that are clipped per frame, so a pixel can be saturated in one frame and not the other, and deep shadows carry mostly sensor noise. Ω_v excludes clipped and near-black pixels in either frame (roughly 75% of pixels survive the mask on average), so the invariance constraint is only applied where both observations are trustworthy. This loss is zero only when the model predicts identical albedo for both images. Minimising it pushes the albedo to be independent of whatever differs between the two illuminations; it says nothing about a component the two frames share.

3.3.3 Explanation Loss

The invariance loss alone is not enough. A degenerate model could satisfy Eq. (3.7) by predicting a flat, constant albedo (the same grey image for every input). To prevent this, we add a companion loss that requires the shading to *explain* the pixel-level difference between the two images.

Under the diffuse model $\mathbf{I} \approx \mathbf{A} \odot \mathbf{S}_d$, the ratio of corresponding pixels in the two images equals the ratio of their shadings wherever the albedo is shared:

$$\frac{\mathbf{I}_1(p)}{\mathbf{I}_2(p)} = \frac{\mathbf{S}_{d,1}(p)}{\mathbf{S}_{d,2}(p)}, \quad \text{if } \mathbf{A}_1(p) = \mathbf{A}_2(p) \quad (3.8)$$

The material term cancels in the ideal diffuse image ratio, making the ratio a useful lighting cue wherever registration is valid and non-diffuse effects are small. Saturation, shadows, and specularities violate that idealisation; the validity mask limits rather than eliminates those failures. We enforce the constraint on the

luminance of both ratios, in the log domain for numerical robustness:

$$\mathcal{L}_{\text{expl}} = \frac{1}{|\Omega_v|} \sum_{p \in \Omega_v} \left| \log \frac{L(\mathbf{I}_1(p)) + \varepsilon}{L(\mathbf{I}_2(p)) + \varepsilon} - \log \frac{L(\mathbf{S}_{d,1}(p)) + \varepsilon}{L(\mathbf{S}_{d,2}(p)) + \varepsilon} \right| \quad (3.9)$$

where $L(\cdot)$ is the Rec. 601 luminance, $\varepsilon = 10^{-3}$, and Ω_v is the same validity mask as in Eq. (3.7). Using luminance makes this term supervise the achromatic *intensity* change between the frames, and only that. A purely chromatic change between the two frames leaves both log-ratios in Eq. (3.9) unchanged and contributes no gradient. Together with $\mathcal{L}_{\text{inv}}$, which a colour shared by both albedos satisfies, nothing in CIAI states where a change in illuminant *colour* must go. On the trained model we measure that the shading explains only 47.9% of the chroma change between paired frames, and that the unpenalised chroma residual (0.074) is as large as the penalised luminance residual (0.076). This is why CIAI is expected to help lightness more than colour (Sect. 5.7). This loss is zero only when the shading ratio matches the image ratio at every valid pixel, which means the shading is absorbing the lighting difference rather than the albedo.

3.3.4 Raw Pairs and the Role of White Balance

MID ships a grey probe in every frame, and the usual pre-processing white-balances each frame by its own probe: the red and blue channels are rescaled so that the probe becomes neutral, while the green channel, and with it the overall intensity, is left unchanged. After this step the two frames of a pair differ in brightness and shadows exactly as before, but their illuminant colours have been equalised.

For the cross-illumination pair we skip this step (the `mid_raw_color_pair` flag). The supervised MID frame stays white-balanced, because the pseudo-ground-truth albedo was derived from the white-balanced stack, while the paired frames keep whatever illuminant colour difference they have, so that $\mathcal{L}_{\text{inv}}$ is asked to be invariant to it. Figure 3.4 shows the two pre-processing lanes.

The size of what is kept should not be overstated. MID’s flash is white, and the colour difference between two raw frames comes only from light bounced off coloured surfaces: measured on the grey probes, the median chromaticity gap between two frames of a scene is 0.030 (Sect. 5.7). Keeping the pairs raw is therefore a choice that preserves a small real colour signal, not a source of strong coloured-illuminant supervision, and our ablations do not isolate its effect from the other components.

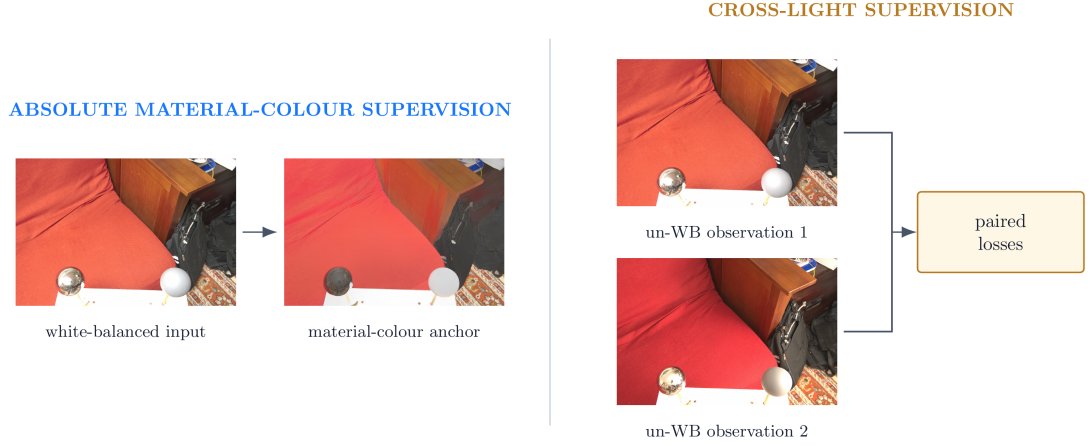


Figure 3.4: The two supervision lanes are built from different pre-processing of the same kind of scene, not different data. The absolute anchor (left) is white-balanced and supervised against the pseudo-ground-truth albedo, which pins $\mathbf{A}$ to a reference colour rather than an arbitrary one. The CIAI pair (right) is left raw, so each observation keeps its own illuminant, including the small colour difference that bounced light introduces between flash directions.

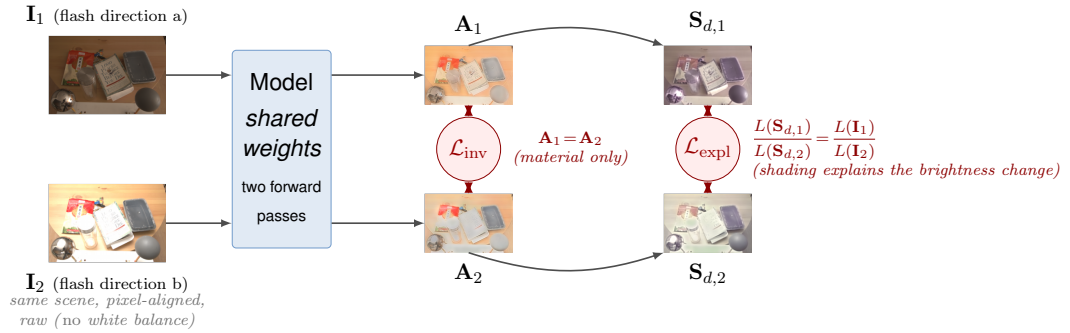


Figure 3.5: Cross-Illumination Albedo Invariance (CIAI), applied only at training time. A pixel-aligned pair of the same MID scene under two flash directions is passed through the same model (shared weights) in two forward passes. $\mathcal{L}_{inv}$ ties the two predicted albedos $\mathbf{A}_1, \mathbf{A}_2$ together, and $\mathcal{L}_{expl}$ requires the luminance ratio of the two shadings $\mathbf{S}_{d,1}, \mathbf{S}_{d,2}$ to match that of the two inputs. The pair is kept raw (Sect. 3.3.4). Thumbnails are this model's own outputs: the two albedos agree closely in lightness and less in colour, which is what the experiments of Chapter 5 measure.

4 | Implementation

This chapter covers the full set of loss terms used during training (Sect. 4.1), the training datasets and how we prepare them (Sect. 4.2), and the optimisation schedule that controls which losses and datasets are active at each stage (Sect. 4.3). Every value in this chapter is taken directly from the training configuration used to produce the final model.

4.1 Loss Terms

Our model is trained with three families of losses: *supervised anchors* that tie the predictions to ground-truth labels on synthetic data, *CIAI losses* that enforce albedo invariance on real multi-illumination pairs, and *physics and regularisation terms* that keep the decomposition consistent. The total loss is a weighted sum; Table 4.1 lists every active term with its weight.

4.1.1 Supervised Anchor Losses

Albedo ($\mathcal{L}_a$, $\mathcal{L}_{\text{dssim}}$, $\mathcal{L}_{\text{msg-a}}$, $\mathcal{L}_{\text{chroma}}$). The core albedo supervision is a masked squared error against the ground-truth albedo $\mathbf{A}^*$, after the ground truth is scaled to the prediction by one least-squares factor per image (albedo is only defined up to scale):

$$\mathcal{L}_a = \frac{1}{|\Omega_m|} \sum_{p \in \Omega_m} \|\mathbf{A}(p) - \mathbf{A}^*(p)\|_2^2 \quad (4.1)$$

where Ω_m excludes invalid pixels (missing labels, clipped regions). This is the dominant *absolute-colour anchor* of the whole system: CIAI constrains albedo to be equal *across* illuminants but says nothing about what the correct colour *is*; without a strong absolute anchor, the invariance constraint alone could be satisfied by grey. Three companions sharpen this anchor. A structural dissimilarity term $\mathcal{L}_{\text{dssim}}$ (DSSIM, weight 0.2) penalises local structure differences. A multi-scale

Table 4.1: Loss terms used in training, with the weights of the final configuration. Supervised anchors require ground-truth labels and are gated to the synthetic datasets that provide them; CIAI terms require multi-illuminant pairs and fire only on MID batches during the CIAI stage; physics terms apply only where their required targets and masks exist. A gradient-decorrelation loss (forcing albedo and shading edges apart) was evaluated and rejected: it assumes albedo and shading edges never coincide, which is false in real scenes, and it erased genuinely shared edges.

Loss term	Symbol	Weight	Applies to
Albedo squared error	$\mathcal{L}_a$	1.0	Hypersim, IV, MID (pseudo-GT)
Albedo DSSIM	$\mathcal{L}_{\text{dssim}}$	0.2	same as $\mathcal{L}_a$
Albedo multi-scale gradient	$\mathcal{L}_{\text{msg-a}}$	0.5	same as $\mathcal{L}_a$
Albedo chroma direction	$\mathcal{L}_{\text{chroma}}$	0.2	same as $\mathcal{L}_a$
Shading SSI (π -domain)	$\mathcal{L}_s$	1.0	Hypersim (GT shading)
Shading multi-scale gradient	$\mathcal{L}_{\text{msg-s}}$	0.5	Hypersim
Diffuse reconstruction	$\mathcal{L}_{\text{recon}}$	1.0	Hypersim
Reconstruction gradient	$\mathcal{L}_{\text{msg-r}}$	0.25	Hypersim
Residual closure + sparsity	$\mathcal{L}_{\text{res}}$	0.5 / 0.02	Hypersim, MID
Material consistency (DINO)	$\mathcal{L}_{\text{mat}}$	0.05	all datasets
Cross-illumination invariance	$\mathcal{L}_{\text{inv}}$	0.5	MID raw pairs
Shading explanation	$\mathcal{L}_{\text{expl}}$	0.25	MID raw pairs
Shading sign prior	$\mathcal{L}_{\text{sign}}$	0.05	all datasets

gradient term $\mathcal{L}_{\text{msg-a}}$ (weight 0.5) matches predicted albedo gradients to ground-truth gradients across four dyadic scales, placing each edge where the ground truth says it belongs. Finally, a *chroma-direction* term (weight 0.2),

$$\mathcal{L}_{\text{chroma}} = \frac{1}{|\Omega_m|} \sum_{p \in \Omega_m} \left\| \frac{\mathbf{A}(p)}{\|\mathbf{A}(p)\|} - \frac{\mathbf{A}^*(p)}{\|\mathbf{A}^*(p)\|} \right\|_1, \quad (4.2)$$

penalises hue and saturation error independently of brightness. This term exists because desaturation is the cheap escape for a pixel-wise loss: predicting a slightly grey version of every colour lowers ℓ_1 risk, and $\mathcal{L}_{\text{chroma}}$ makes that escape expensive.

Shading SSI ($\mathcal{L}_s$, $\mathcal{L}_{\text{msg-s}}$). Shading is only defined up to an arbitrary global scale: a dimmer light with a brighter material explains the same image. The renderer’s absolute light intensity varies freely between scenes, so a naive regression to ground-truth shading would spend most of its gradient fighting a quantity that is not identifiable. The standard remedy, introduced for the analogous scale ambiguity in monocular depth estimation (Ranftl et al., 2022) and adopted for shading by

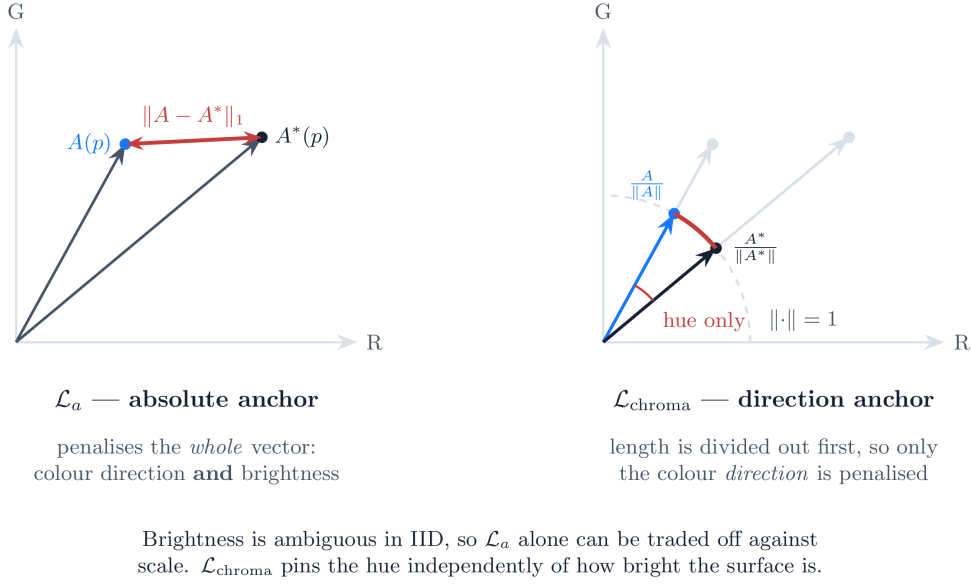


Figure 4.1: Eq. (4.1) and Eq. (4.2) differ only in whether the vectors are normalised first, but that changes what error each one can see. $\mathcal{L}_a$ (left) is the raw distance between the predicted and reference albedo vectors, so it conflates hue error with brightness error — a prediction can lower this loss by getting closer in scale without getting the colour right. $\mathcal{L}_{\text{chroma}}$ (right) divides out the length of both vectors before comparing them, so only the angle between them — the hue — is penalised. This is the same failure this thesis’s corrected MID metrics are built to catch at evaluation time (Sect. 5.2): a term that cannot distinguish scale from colour will always be satisfiable by discarding colour.

Careaga and Aksoy (2023), is a *scale-and-shift-invariant* (SSI) loss: before the error is computed, the prediction is aligned to the ground truth with the per-image scalar scale $\hat{s}$ and shift $\hat{t}$ that minimise the squared error, a two-parameter least-squares problem with a closed-form solution, so the alignment adds no optimisation difficulty. The aligned prediction is then penalised with a masked mean-squared error in the bounded $\boldsymbol{\pi}$ domain:

$$\mathcal{L}_s = \frac{1}{|\Omega_d|} \sum_{p \in \Omega_d} (\hat{s} \boldsymbol{\pi}(p) + \hat{t} - \boldsymbol{\pi}^*(p))^2, \quad (\hat{s}, \hat{t}) = \arg \min_{s, t} \sum_{p \in \Omega_d} (s \boldsymbol{\pi}(p) + t - \boldsymbol{\pi}^*(p))^2 \quad (4.3)$$

Ω_d is the *diffuse mask*: shading ground truth exists only on Hypersim, so this loss is gated to Hypersim batches. This supervises the relative shading pattern (shadows, gradients) without being confused by per-scene brightness. A multi-scale gradient companion (weight 0.5) sharpens shading boundaries. The SSI alignment is enabled

after a 3,000-iteration warm-up, during which the un-aligned loss stabilises the output range first.

Diffuse reconstruction ($\mathcal{L}_{\text{recon}}, \mathcal{L}_{\text{msg-r}}$). Because the residual is analytic ($\mathbf{R} = (\mathbf{I} - \mathbf{A} \odot \mathbf{S}_d)_+$), a naive reconstruction loss $\|\mathbf{A} \odot \mathbf{S}_d + \mathbf{R} - \mathbf{I}\|$ would be identically zero wherever the diffuse product undershoots the image, which is a tautology that provides no gradient. We therefore supervise the *diffuse product* directly:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\Omega_d|} \sum_{p \in \Omega_d} \|\mathbf{A}(p) \odot \mathbf{S}_d(p) - \mathbf{A}^*(p) \odot \mathbf{S}_d^*(p)\|_1 \quad (4.4)$$

The term is enabled only for Hypersim, where the target is the measured ground-truth diffuse product. It is a *two-sided* coupling: both predicted factors must jointly explain $\mathbf{A}^* \odot \mathbf{S}_d^*$. InteriorVerse and MID do not activate this loss in the implemented configuration; those batches are supervised by their available albedo, material, residual, sign, and CIAI terms instead. This masking is important: defining an input-derived target for datasets without measured shading would change the trained objective.

Residual closure ($\mathcal{L}_{\text{res}}$). The analytic residual creates an escape hatch: the model could dump hard-to-explain image content (for example a saturated red surface) into $\mathbf{R}$ instead of the albedo. We close it with a supervised residual block (weight 0.5): on Hypersim the residual target is the true non-diffuse component of the rendering, and on MID the target is $\mathbf{R}^* = 0$ away from specular pixels (physically true for the bounced-flash captures), so the loss there penalises *any* residual energy. A small sparsity term (weight 0.02) additionally pushes $\mathbf{R} \rightarrow 0$ inside the valid mask, so the diffuse product rather than the residual must explain the image.

4.1.2 CIAI Losses

The two CIAI losses $\mathcal{L}_{\text{inv}}$ (weight 0.5) and $\mathcal{L}_{\text{expl}}$ (weight 0.25) are defined in Sect. 3.3. They are computed by running the network a second time on the paired frame $\mathbf{I}_2$ within the same training step (a Siamese second forward with shared weights) and are masked by the intersection of the loss mask, the material-invariance mask, and the per-pair HDR validity mask. They fire on batches that contain MID raw pairs, i.e. from the CIAI stage onwards (Sect. 4.3).

4.1.3 Physics and Regularisation Losses

Shading sign prior ($\mathcal{L}_{\text{sign}}$). The analytic residual is clamped non-negative, so any pixel where the diffuse product *overshoots* the image is un-modelled negative energy, which is something the decomposition cannot represent. The sign prior penalises exactly that overshoot:

$$\mathcal{L}_{\text{sign}} = \frac{1}{|\Omega_m|} \sum_{p \in \Omega_m} \|\max(\mathbf{A}(p) \odot \mathbf{S}_d(p) - \mathbf{I}(p), 0)\|_1 \quad (4.5)$$

The intuition is “anything that brightens the image beyond the light must come from the material, not the shading”: the term pushes brightening texture out of the shading and into the albedo, complementing CIAI (which pushes the change in illumination between two frames out of the albedo and into the shading).

Material consistency ($\mathcal{L}_{\text{mat}}$). The frozen DINOv2 tokens provide a label-free material signal: two patches with very similar tokens are very likely the same material. The material-consistency term (weight 0.05) penalises albedo chroma differences between patch pairs whose DINOv2 similarity exceeds a threshold, gently flattening the albedo *within* a material while leaving material boundaries untouched. Since the tokens are computed anyway for the encoder, this supervision is essentially free.

4.2 Training Data

Three datasets participate in training, each with a distinct role. In this chapter, *dataset* means a collection from which training samples are drawn. It does not mean an evaluation procedure. Chapter 5 uses held-out data together with a fixed protocol and metric; that combination is a *benchmark*. MID appears in both roles, but its training and test scenes are disjoint. They are existing public corpora and carry the two-stage schedule of Sect. 4.3: Hypersim and InteriorVerse supply supervised albedo, and MID supplies the real cross-illumination pairs that CIAI trains on.

4.2.1 Hypersim

Hypersim (Roberts et al., 2021) is a large photorealistic synthetic dataset of 461 indoor scenes generated with a physically-based renderer, providing ground-truth albedo (diffuse reflectance) and diffuse illumination for every image. It is the primary source of supervised signal: albedo, shading, reconstruction, and residual

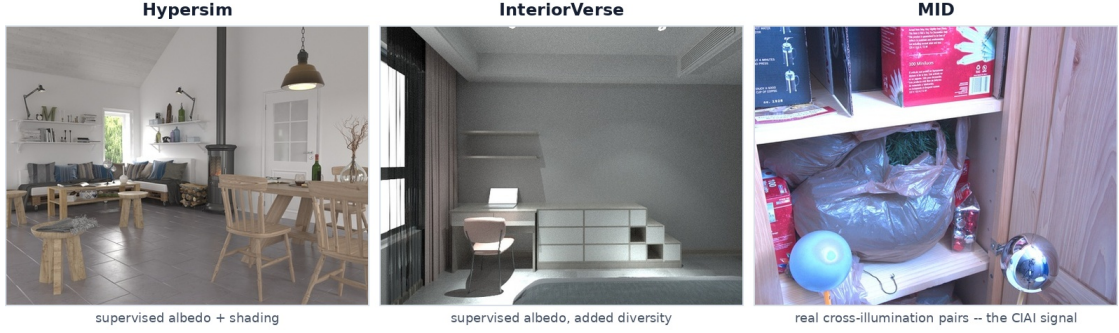


Figure 4.2: *Hypersim and InteriorVerse are both synthetic and both supply supervised albedo; MID is real and is the only source of the paired cross-illuminant observations CIAI is trained on. No single dataset provides everything the model needs.*

targets all come from Hypersim labels. The full release is large (multiple terabytes of EXR renders); under a local storage budget of 500 GB we download and train on approximately one quarter of the scenes, sampled across the scene split so that the material and layout diversity is preserved. This subset is sufficient because Hypersim’s role is to anchor absolute albedo colour and supply shading ground truth, not to cover every scene; the cross-illuminant signal that the thesis depends on comes from the real MID pairs (Sect. 2.4.2), not from Hypersim. We use the standard scene split within the subset and hold out a validation set for monitoring and checkpoint selection.

4.2.2 InteriorVerse

InteriorVerse (Zhu et al., 2022a) is a rendered indoor dataset with ground-truth material, geometry, and spatially varying lighting annotations. It enters training as *albedo-supervision diversity*: more scenes, layouts, and materials for the absolute-colour anchor. Its shading is defined only implicitly ($\mathbf{S}_d^* = \mathbf{I}/\mathbf{A}^*$), so it is excluded from both shading-GT and diffuse-reconstruction losses in the implemented configuration. InteriorVerse contributes albedo supervision and material diversity, rather than a separately measured diffuse-image target.

4.2.3 MID: Real Multi-Illuminant Pairs

MID (Murmann et al., 2019) provides the cross-illumination training signal: over 1,000 real indoor scenes, each captured 25 times with a flash bounced in a different direction, pixel-aligned, with the 30 test scenes held out (Sect. 2.4.2). Each MID

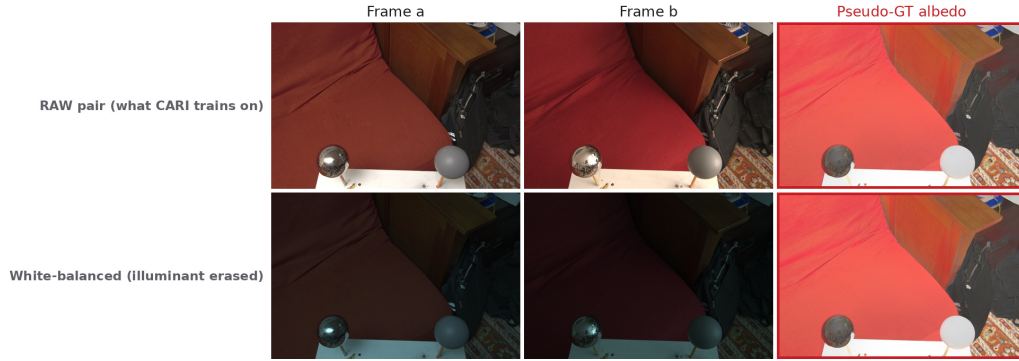


Figure 4.3: *MID preprocessing used during training; this is a data-loading diagram, not an evaluation protocol. The supervised path uses a white-balanced primary frame and the dataset pseudo-ground-truth albedo to anchor absolute material colour on real photographs. The CIAI path keeps two aligned frames raw, without white balance, and masks pixels that are clipped or too dark in either exposure before applying the cross-illumination losses. This split keeps the albedo target stable while preserving the illuminant chromaticity change that CIAI needs as supervision.*

training sample contributes two things:

(i) *A supervised primary frame.* The MID Intrinsic extension of Careaga and Aksoy (2023) supplies a dense pseudo-ground-truth albedo per scene, derived from the white-balanced multi-illumination stack; it is not part of the original MID release. The white-balanced primary frame is supervised against this pseudo-GT with the standard albedo losses, which anchors absolute colour on real photographs.

(ii) *A raw cross-illumination pair.* A second frame of the same scene, under a different bounce direction, is loaded *raw*, without white balance, together with a per-pair HDR validity mask that drops pixels clipped or near-black in either frame. The CIAI losses operate on this pair. Skipping white balance here (the `mid_raw_color_pair` flag, Sect. 3.3.4) keeps the small bounce-colour difference between the two frames; the brightness and shadow change, which is the main signal MID provides, is present either way. An earlier configuration added random global tints to the pair frame to create colour differences synthetically; it did not help in our early tests and was dropped. Figure 4.3 summarises this preprocessing split: the white-balanced primary frame provides the supervised colour anchor, while the raw paired frames preserve the illuminant change used by CIAI.

4.3 Optimisation Schedule

Training proceeds in two stages, as illustrated in Figure 4.4.

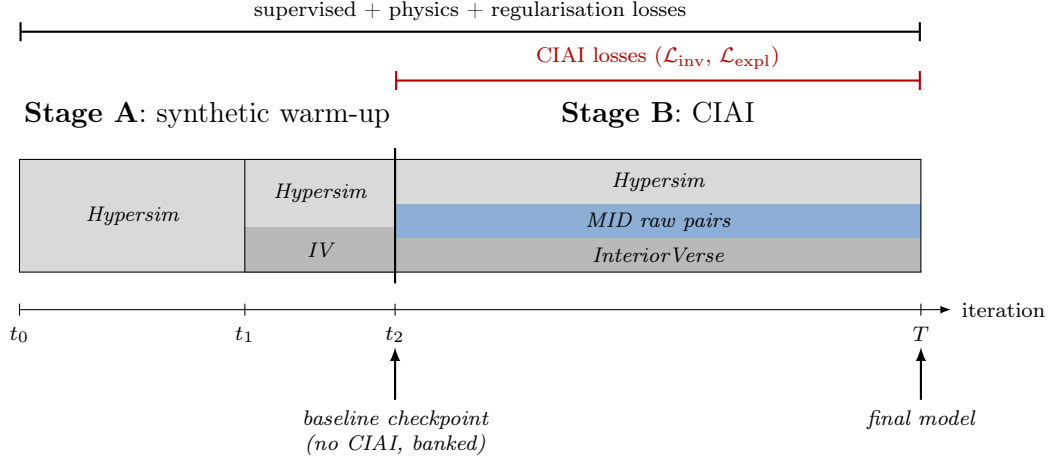


Figure 4.4: Two-stage training schedule. Stage A establishes the supervised baseline on synthetic data (Hypersim, later joined by InteriorVerse). Stage B resumes from the Stage-A checkpoint and adds MID raw pairs with the CIAI losses, keeping the synthetic majority as the absolute-colour anchor. The Stage-A checkpoint doubles as the no-CIAI baseline in the ablation study (Sect. 5.6).

Stage A: synthetic warm-up. The model first trains on Hypersim alone, after which InteriorVerse joins the mix. Only the supervised, physics, and regularisation losses are active. This stage gives the model absolute albedo colour and scale *before* any invariance constraint is applied. The ordering matters because an invariance loss applied to an unanchored model can be satisfied by collapsing towards grey; on an anchored model the supervised losses keep penalising that collapse while the invariance loss acts. The Stage-A checkpoint is banked as the no-CIAI baseline used throughout Chapter 5.

Stage B: CIAI. Training resumes from the Stage-A checkpoint with the CIAI configuration: the data mix becomes Hypersim / MID / InteriorVerse (with MID and InteriorVerse each roughly matching Hypersim’s share), the raw cross-illumination pairs enter through MID, and $\mathcal{L}_{\text{inv}}/\mathcal{L}_{\text{expl}}$ activate. The synthetic majority keeps re-anchoring absolute colour throughout, preventing drift while the invariance constraint acts on the MID pairs. The end-of-schedule checkpoint is the final model reported in Chapter 5. We describe the schedule here by stage rather than by a specific iteration count, so that every configuration compared in Chapter 5 is held to the same curriculum and the same total training budget.

Optimisation parameters. We use Adam with a base learning rate of 2×10^{-4} and cosine annealing to 10^{-7} over the training horizon; the frozen DINOv2 encoder

is excluded from the optimiser entirely. Batches of 4 crops at 512×512 are accumulated over 8 steps (effective batch size 32), with gradient-norm clipping at 1.0. All training runs on a single 24 GB NVIDIA Quadro RTX 6000.

The reported configuration. The model reported as “Ours” is this architecture trained under the two-stage schedule above to 40k iterations (Stage A to 19k, Stage B from 19k to 40k): Stage A establishes absolute albedo colour and shading structure from Hypersim and InteriorVerse, and Stage B adds the real MID raw cross-illumination pairs and activates the cross-illumination losses $\mathcal{L}_{\text{inv}}$ and $\mathcal{L}_{\text{expl}}$ together with the wide colour skip. Because the cross-illumination terms require a *second* forward pass on the paired frame, the CIAI stage holds two graphs in memory; on the 24 GB card this caps the per-step crop count, which the gradient accumulation above compensates for. The frozen DINOv2-Large encoder (304 M parameters) is kept in evaluation mode throughout, so only the ~ 18.5 M decoder-and-head parameters receive gradients, which is the main reason the whole pipeline fits on a single commodity GPU.

5 | Experiments

This chapter reports results on the four benchmarks of Sect. 2.4, after the experimental setup in Sect. 5.1.

The order of the benchmark sections is deliberate, because the benchmarks do not carry equal weight and presenting them as a flat list would suggest that they do. We begin with MID (Sect. 5.2), the only benchmark that measures the property this thesis is about: whether the predicted albedo of a real surface stays put when the light on it changes. We then turn to IIW (Sect. 5.3), the field’s long-standing perceptual benchmark. Its human judgements validly test relative reflectance ordering, but WHDR is blind to absolute albedo and hue. It is therefore complementary evidence rather than the deciding measure for coloured-illuminant leakage. That limitation is the reason MAW exists, so MAW follows (Sect. 5.4): it compares predictions against physically measured albedo on real surfaces and scores colour error directly, which is the closest thing the field has to an objective verdict on material colour. ARAP comes last (Sect. 5.5) because it is synthetic and therefore easiest to satisfy, but it is also the only benchmark with dense ground-truth albedo *and* repeated renderings of one scene under different lights, so it is the one place where constancy and accuracy can be measured on the same pixels.

All four benchmarks score the *albedo*, and that is deliberate rather than an oversight. The question this thesis asks is whether a surface’s recovered colour survives a change of illuminant, so the albedo is the quantity under test; the shading is the layer into which the illuminant is displaced, and it is exercised qualitatively by factor-space edits in Chapter 6. We therefore do not report SAW (Careaga and Aksoy, 2023), which scores shading annotations: a good SAW score would say nothing about whether material colour is stable, which is the claim being made. This does mean that a decomposition could in principle satisfy every benchmark here while producing an implausible shading layer, and we do not rule that out. The application chapter demonstrates controllability, not a ground-truth shading evaluation.

Sect. 5.6 then isolates what CIAI itself contributes, and Sect. 5.7 measures why colour stability improves much less than lightness.

5.1 Experimental Setup

Model used in the results. **Ours** is the complete CIAI configuration of Chapter 3: both cross-illumination losses and the colour path active, trained to 40k iterations. It is Row 4 of the ablation in Sect. 5.6, and it is the model meant wherever the text says “our model”. The other ablation rows appear only in Sect. 5.6.

Qualitative figures. The tables report the model above. Several qualitative figures were rendered before the reported model was fixed, with later checkpoints of the same architecture trained for 20k further steps from it, and will be regenerated once its weights are available. The decomposition figures of Chapter 3 and Chapter 6 and the MAW and ARAP examples use a later checkpoint of the same model; the MID, IIW, resolution and chroma figures use one continued with additional albedo regularisers. Their predictions look alike, but the numbers printed inside them, where present, are not the reported ones.

Baselines and how scores are obtained. The benchmark tables compare against the same roster of methods, and follow the convention used by (Careaga and Aksoy, 2024): a horizontal rule separates scores we *run locally* from scores we *cite* from the literature. We run Marigold-IID Appearance and Lighting (Ke et al., 2025), CRefNet (Luo et al., 2024), the Ordinal Shading stage of CD-IID (Careaga and Aksoy, 2023), the full five-stage CD-IID cascade (Careaga and Aksoy, 2024), and RGB→X (Zeng et al., 2024) locally under our exact pipeline (same output-space normalisation and splits, identical inference resolution per benchmark). The last two were integrated after the first submission of this work; they are evaluated on MID here, and the older baselines are cited from their publications and appear below the rule in each table. Our own constancy metrics (C_{mat} , C_{arap}) are not reported by those methods and are marked n/a; a cell marked “-” is simply one for which no value is available. Marigold-IID Lighting is trained on Hypersim, like our model, which makes it the most directly comparable external method.

Which published table each benchmark cites. The published comparison point differs per benchmark, chosen so that the cited protocol matches ours exactly rather than the nearest available number. MAW cites CD-IID Table 1: both the MAW authors’ own evaluation code and our local harness apply the identical scaling and metrics. ARAP cites Ordinal Shading Table 1, *not* CD-IID Table 2: CD-IID modifies the ARAP scene set (dropping redundant illuminations and adding scenes from the MIST dataset), a set we cannot reproduce, whereas Ordinal Shading’s table uses the original Bonneel et al. scenes (the same 52-scene set used throughout this chapter). IIW cites Ordinal Shading Table 2 for its zero-shot WHDR figure,

alongside the fuller reproduced roster of (Wang et al., 2025). MID has no external citation on any axis; it is a ground-truth-free constancy metric that no other published work reports, and MIDIntrinsics is training data for both this model and CD-IID, which additionally disqualifies it as a zero-shot benchmark for either.

Methods compared. Table 5.1 summarises the methods along the axes that condition a fair reading of the benchmark numbers: architecture type, training data, parameter count, runtime, and native output space. Two points matter for interpretation. First, training data determines what counts as zero-shot. Our model trains on Hypersim, MID, and InteriorVerse, so it shares one source with each Marigold variant (Hypersim with Marigold-Lighting and InteriorVerse with Marigold-Appearance), which makes both directly comparable rather than only one. It is zero-shot on ARAP, MAW, and IIW, but in-domain on MID (it trains on the MID training split, disjoint from the test scenes), which is why the MID constancy metric is deliberately ground-truth-free (Sect. 5.2). CRefNet trains on the IIW training split, so its IIW WHDR is a non-zero-shot result (Sect. 5.3). Second, our model trains only about 18.5M parameters. The DINOv2-L encoder (304M) is frozen and excluded from the optimiser, so the trainable count is far lower than for the diffusion baselines ($\sim 1290\text{M}$) or the five-stage CD-IID cascade (442.8M, every stage trained), an efficiency the constancy results should be read against. On *total* parameters the gap is much smaller — our frozen backbone puts us at 322.9M, only $1.4\times$ below CD-IID and above CRefNet’s 66.6M — so the saving is specifically in weights that had to be optimised, not in deployed model size.

Output-space normalisation. Marigold-IID’s two variants use different output conventions: the Appearance variant outputs sRGB albedo, while the Lighting variant outputs linear albedo, shading, and residual layers. Before computing any metric, each prediction is converted to the corresponding linear albedo space; for sRGB predictions this applies the transfer function $A_{\text{lin}} = A_{\text{sRGB}}^{2.2}$. This correction matters because gamma encoding can artificially desaturate an albedo estimate and make it appear more constancy-correct than it actually is. All results in this chapter use this corrected normalisation.

Inference resolution. Inference resolution is fixed per benchmark rather than shared uniformly. MID, IIW, and ARAP use an aspect-preserving maximum side length of 1280 pixels; measured directly on the full ARAP set, 1280 and 1500 pixels give effectively identical scores (every metric moves by less than 1%, in mixed directions, consistent with measurement noise rather than a real effect), so 1280 is kept throughout for consistency and lower memory use. Ordinal Shading (Careaga and Aksoy, 2023) instead resizes each ARAP image to a content-adaptive resolution

Table 5.1: *Methods compared by architecture, training data, model size, inference cost, and native output. Runtime is the median of five timed runs after warm-up at 512×512 on one NVIDIA Quadro RTX 6000, using the same local interface; Marigold uses its recommended four denoising steps. Trainable parameters exclude frozen weights. A dash denotes a quantity not reported by the paper and not measured under our protocol. ReasonX’s MLLM is a training-time judge, not the deployed intrinsic predictor.*

Method	Type	Training data	Train (M)	Total (M)	s/image	Output
Ours	DINOv2-L (frozen) + DPT, regression	Hypersim + M1D + InteriorVerse	18.5	322.9	0.148	linear albedo
Marigold-App (Ke et al., 2025)	Stable-Diffusion latent diffusion	InteriorVerse	~1290	1290	0.805	sRGB albedo + material
Marigold-Light (Ke et al., 2025)	Stable-Diffusion latent diffusion	Hypersim	~1290	1290	0.982	linear albedo + shading + residual
CD-I1D (Careaga and Aksoy, 2024)	Five-stage MiDaS-style cascade	8 synthetic sets + M1DIntrinsics; Hypersim for shading	442.8	442.8	0.254	linear albedo + RGB shading + residual
Ordinal Shading (Careaga and Aksoy, 2023)	CD-I1D stage 1; ordinal MiDaS shading	CGIntrinsics + Hypersim + OpenRooms + GTA + M1D	~337	~337	–	grayscale shading + implied albedo
PIE-Net (Das et al., 2022)	Edge-guided hybrid CNN	synthetic outdoor; MIT/I1W fine-tunes	204.1	204.1	–	reflectance + shading
CRefNet (Luo et al., 2024)	Decoder-sharing SwinV2 transformer	CGIntrinsics + I1W (+ N1D-Net pseudo-shading)	66.6	66.6	0.447	reflectance + shading
ReasonX (Dirik et al., 2025)	GRPO fine-tuning with InternVL2.5-4B judge	10k COCO RGB; judge: Hypersim + InteriorVerse	–	–	–	base-dependent intrinsic maps



Figure 5.1: *Effect of inference resolution on a MAW scene using the full-CIAI model. The input photograph is shown on the left and the measured albedo on the right; the middle columns show predictions at different maximum side lengths. Each panel repeats the same black-sofa-armrest crop in the upper right with a red border to make blur differences easier to compare. Fine structure improves substantially from 512 to 1280 pixels, then largely plateaus. We therefore use the common 1280-pixel inference cap for every locally run MAW method; the official scorer subsequently reduces predictions to 320×240 before measuring the metrics.*

R_0 (Miangoleh et al.’s edge-density criterion), capped at 1500 pixels; since R_0 is computed per image and not reproducible from the published protocol alone, no fixed resolution can replicate it exactly, and the 1280-vs-1500 measurement above shows that choice is immaterial to our own scores regardless. MAW also uses the common 1280-pixel inference cap. Its public scorer then resizes every prediction and region mask to 320×240 before computing the metrics, so the reported numbers are determined at that fixed scoring scale. Every locally run method, including the baselines, is evaluated at the same resolution within each benchmark. Figure 5.1 shows the qualitative plateau this is chosen from: raising the full-CIAI prediction from 1280 to 1792 or 2560 changes little visually, whereas 512 visibly softens fine structure at the resolutions where that matters (MID, IIW).

Benchmarks at a glance. Figure 5.2 previews the four evaluation benchmarks and the kind of ground truth each provides, which determines the metric it supports (Sect. 2.4). Figure 5.3 shows the qualitative comparison format used throughout this chapter: for a single in-the-wild image, the input and each method’s predicted albedo side by side.

5.2 MID: Multi-Illuminant Constancy

MIDIntrinsics (Murmann et al., 2019) photographs the same real room 25 times, moving only the light, so it isolates the exact variable this thesis is about. Because it ships no independently measured dense albedo, it cannot tell us whether a prediction is *correct*; it can only tell us whether a prediction is *stable*. That makes it a constancy diagnostic rather than an accuracy benchmark, and we report it as such. We use the held-out 30-scene test split. The two scores follow the reflectance-

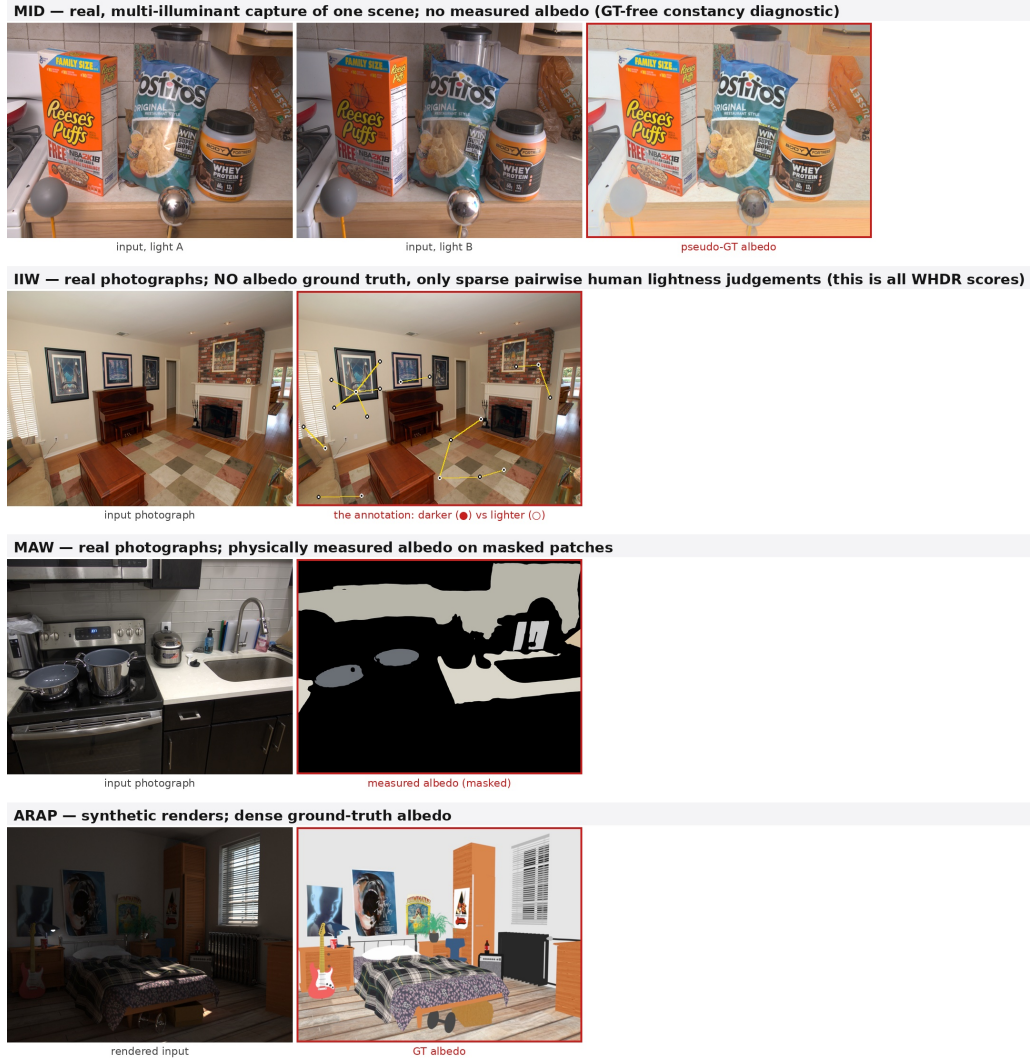


Figure 5.2: The four evaluation benchmarks and the ground truth each provides, in the order the chapter presents them. **MID:** real multi-illuminant captures of one scene, with no measured albedo; which is why it supports a ground-truth-free constancy diagnostic and nothing stronger. **IIW:** real photographs with no albedo ground truth at all. The second panel overlays the annotation that does exist: sparse pairwise human lightness judgements, each a pair of points one of which was called darker (●) than the other (○). These sparse ordinal comparisons define what WHDR scores; they are informative about relative lightness but not RGB colour. **MAW:** real photographs with physically measured albedo on masked patches. **ARAP:** synthetic renders with dense ground-truth albedo.

consistency setting of (Kinoshita and Kiya, 2021) but are computed directly on

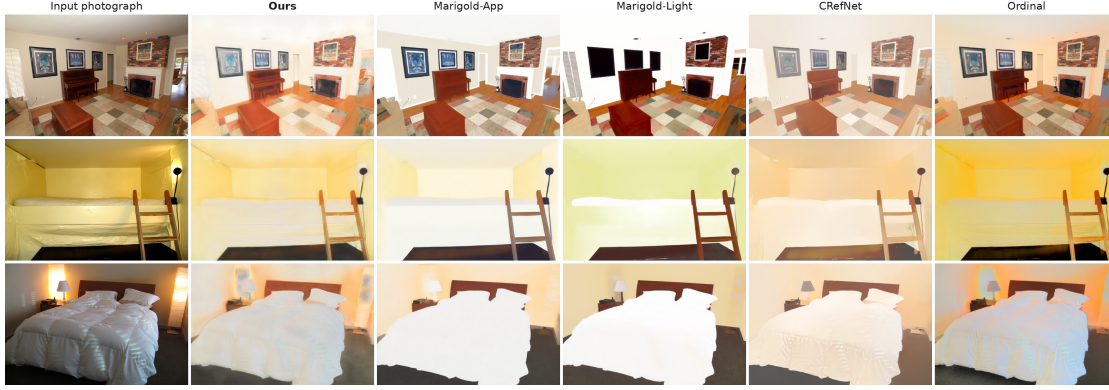


Figure 5.3: Qualitative albedo comparison on in-the-wild photographs (IIW), all methods run under the identical evaluation protocol. Each panel is scale-normalised by its own 99th percentile, because albedo is defined only up to a global scale and each method fixes that scale differently; beyond this, no colour correction is applied. The comparison is deliberately shown, but it cannot be adjudicated: IIW supplies no albedo ground truth (Figure 5.2), so an albedo that looks “cleaner” cannot be distinguished by eye from one that has simply had its colour removed. The diffusion baselines whiten these interiors considerably more aggressively than we do, and on this evidence alone one might read that as a better decomposition. The MID pseudo-GT calibration in Table 5.2 shows the ambiguity: Marigold-App and CRefNet retain 72.8% and 48.4% of the reference chroma spread, against our 94.1%. This figure shows method behaviour; it cannot establish correctness on IIW.

MID’s illuminant groups.

Both scores rest on the same fact: within one scene, the material at a given pixel is the same in every one of the 25 frames, so any variation in the predicted albedo across frames is error by definition. Let $k = 1, \dots, N$ index the illuminant conditions of a scene and let m index the material regions given by the scene’s segmentation.

The first score asks whether the material’s *lightness* holds still. Writing $\mu_{m,k}$ for the mean luminance of the predicted albedo over material m in frame k , we take the coefficient of variation across frames and average it over materials,

$$C_{\text{mat}} = \frac{1}{|M|} \sum_{m \in M} \frac{\text{std}_k(\mu_{m,k})}{\text{mean}_k(\mu_{m,k})}. \quad (5.1)$$

Dividing by the mean makes the score scale-free, which matters because albedo is only defined up to a global scale.

The second score asks whether the material’s *colour* holds still, which is the axis a coloured lamp actually attacks. We reduce each material in each frame to its

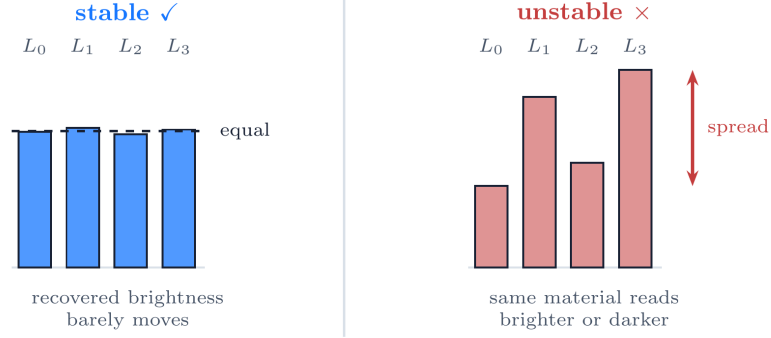


Figure 5.4: C_{mat} is the coefficient of variation of one material’s mean luminance across the scene’s illuminant conditions. A stable prediction gives four nearly equal bars; a prediction whose brightness tracks the lamp gives four bars with visible spread, which is exactly the quantity the standard deviation in Eq. (5.1) measures.

two chromaticity ratios $r_{m,k} = \bar{A}_{m,k}^R / \bar{A}_{m,k}^G$ and $b_{m,k} = \bar{A}_{m,k}^B / \bar{A}_{m,k}^G$. Normalising by the green channel removes brightness and leaves hue, so a model that lets a warm lamp tint its albedo will show drift in $r_{m,k}$ and $b_{m,k}$ even if its lightness is stable.

A pooled chroma spread is not a stability metric. The first version of this evaluation, in early drafts of this work, pooled these ratios over every material and every frame and reported their combined spread,

$$\text{Cast}_{\text{RMS}} = \sqrt{\text{Var}(r_{m,k}) + \text{Var}(b_{m,k})}, \quad (5.2)$$

where the variance runs over all (m, k) pairs jointly. This was our own choice, not a protocol taken from the literature, and it is unsound. By the law of total variance the pooled variance splits into

$$\underbrace{\text{Var}_{m,k}(r_{m,k})}_{\text{pooled}} = \underbrace{\mathbb{E}_m[\text{Var}_k(r_{m,k})]}_{\text{within: drift across illuminants}} + \underbrace{\text{Var}_m(\mathbb{E}_k[r_{m,k}])}_{\text{between: chroma spread across materials}}. \quad (5.3)$$

Only the first term is drift. The second is the chromatic diversity of the scene, and because it is computed from the *predicted* albedo, a model that pulls distinct materials toward a common hue lowers Eq. (5.2) without becoming any more stable. In the ablation of Sect. 5.6 the pooled score ranked the visibly desaturated configuration best. We record the mistake because it is easy to make: any stability score computed on predictions alone can be improved by discarding colour, and must be read next to an accuracy score.

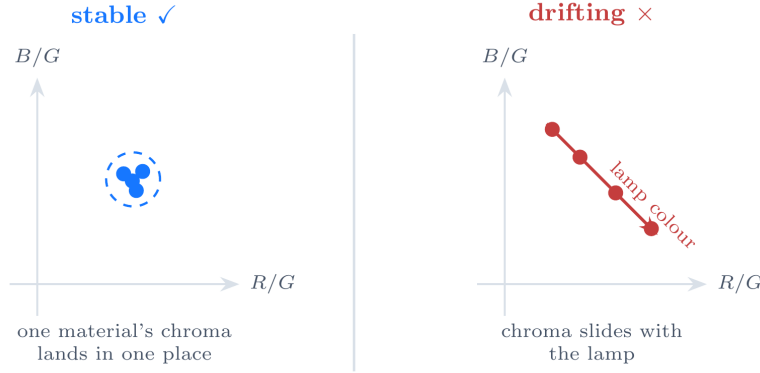


Figure 5.5: Plotting one material’s per-illuminant chromaticity $(r_{m,k}, b_{m,k})$ as a scatter makes the two failure modes visually distinct: a stable material clusters tightly, while a material whose colour tracks the lamp strings out along the illuminant-colour direction. δ_m (Eq. (5.4)) is the radius of that cluster.

Reported metrics. We report hue stability from the first term of Eq. (5.3) only. Write the per-material drift and the between-material chroma spread of a prediction A as

$$\delta_m = \sqrt{\text{Var}_k(r_{m,k}) + \text{Var}_k(b_{m,k})}, \quad S(A) = \sqrt{\text{Var}_m(\bar{r}_m) + \text{Var}_m(\bar{b}_m)}, \quad (5.4)$$

with $\bar{r}_m = \mathbb{E}_k[r_{m,k}]$ and $\bar{b}_m = \mathbb{E}_k[b_{m,k}]$. The constancy score is the mean drift measured against the chroma scale the model actually uses,

$$\text{Cast}_{\text{rel}} = \frac{\frac{1}{|M|} \sum_{m \in M} \delta_m}{S(A)}, \quad (5.5)$$

a coefficient of variation in chroma. The normalisation is necessary because δ_m is an absolute quantity: a model that carries twice as much albedo chroma has twice as much room to drift in absolute units, so comparing raw δ_m across models with different colourfulness penalises the more colourful one for being colourful. Cast_{rel} instead asks the question that matters perceptually: whether two materials remain distinguishable when the lighting changes. Being a ratio, it has one blind spot: a change that lowers the drift and the model’s own chroma by the same fraction leaves it unchanged.

Accuracy beside stability. Stability alone can never be reported on its own, because a constant predictor attains it: a model that emits flat grey albedo is *perfectly* stable and scores zero on every drift measure. Every stability number in this chapter is therefore read next to an accuracy number. The external accuracy

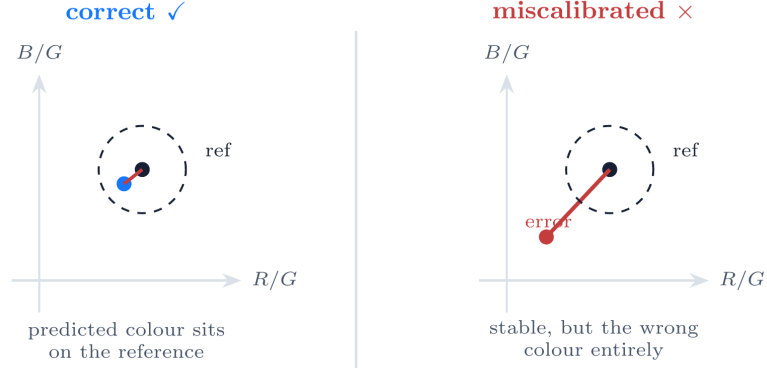


Figure 5.6: $Cast_{rel}$ alone cannot see this failure: a prediction can be perfectly clustered (stable) yet sit on entirely the wrong point in chroma space. $Chroma_{err}$ is the distance from that cluster to the pseudo-GT reference, so it is zero only when the stable colour is also the right one.

measure is MAW (Sect. 5.4): CIEDE2000 error against physically measured albedo, comparable with published tables. It is not sufficient here on its own, for two reasons. It scores different photographs, one illumination each, so it cannot tell whether the very frames on which stability was measured were washed out; and it does not reliably register desaturation. CRefNet, which keeps less than half of the reference chroma spread on MID, scores a MAW ΔE of 3.970, better than our model’s 4.155, although MAW does separate our own desaturated ablation row from its colour-preserving counterpart (4.631 against 4.155). We therefore add an in-domain accuracy measure on the same MID frames, the mean per-material chromaticity error against the pseudo-ground truth,

$$Chroma_{err} = \frac{1}{|M|} \sum_{m \in M} \|(\bar{r}_m, \bar{b}_m) - (\bar{r}_m^*, \bar{b}_m^*)\|, \quad (5.6)$$

where $(\bar{r}_m^*, \bar{b}_m^*)$ is the pseudo-GT chromaticity of material m ; lower is better. This is an ordinary accuracy measure, restricted to chromaticity, not a new metric. Its limitation is the reference itself: the pseudo-ground truth is derived from MID’s capture stack, and both our model and CD-IID train on MIDIntrinsics.

For diagnosis only, we also quote the ratio of the between-material chroma spread of a prediction to that of the reference,

$$\text{spread ratio} = \frac{S(A)}{S(A^*)}, \quad (5.7)$$

computed as the ratio of aggregate spreads over the test split. It is not an accuracy measure (a permutation of hues preserving the spread would score 1), but it makes desaturation visible as a value well below 1 (Figure 5.7).

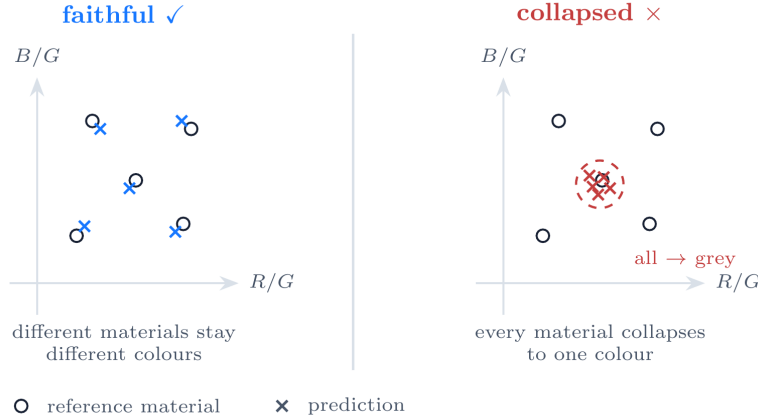


Figure 5.7: The spread ratio compares the chroma spread of the predicted materials with that of the reference. A faithful prediction keeps different materials at different colours (ratio near 1); a desaturated one collapses them towards grey (ratio well below 1), which would also make it look stable.

C_{mat} and Cast_{rel} are lower-is-better and both vanish for a perfectly illuminant-invariant prediction. They are reported separately, rather than combined, because they can move in opposite directions: C_{mat} is a lightness score and Cast_{rel} a hue score, and a configuration may stabilise one while disturbing the other.

This is deliberately an in-domain diagnostic: our method trains on MID, and so does CD-IID (Careaga and Aksoy, 2024). We train on the official MID training split and hold out its 30-scene test split, which is the split evaluated throughout this chapter. CD-IID’s description of its training data names the multi-illumination dataset without stating a split, so we cannot verify which scenes it saw; that is a limit on what any comparison against it can claim, not an allegation. The table therefore compares locally evaluated models under one pipeline and makes no zero-shot claim.

Table 5.2 is not a victory on any single column, and we report the losses first, because the paired bootstrap of Table 5.3 makes them unambiguous.

Our model does *not* lead hue stability. Every external method except Ordinal Shading has a lower Cast_{rel} than our 0.425, and the paired difference is significant for four of the five. CD-IID leads the column at 0.306.

It does not lead chroma accuracy either. Its $\text{Chroma}_{\text{err}}$ of 0.129 is second to CD-IID’s 0.090, and that gap is significant (+0.039 [+0.022, +0.057]). CD-IID reaches better accuracy *and* better hue stability while keeping its colour (spread ratio 0.944), so low hue drift is not only reachable by desaturating.

On lightness, our model is significantly more stable than CD-IID, both Marigold variants and Ordinal Shading, level with CRefNet (+0.007 [−0.004, +0.019]), and

Table 5.2: *Stability and in-domain accuracy on the held-out MIDIntrinsics test split (Murmman et al., 2019), 30 scenes; brackets are 95% percentile-bootstrap intervals over scenes. C_{mat} (Eq. (5.1)) is lightness stability and Cast_{rel} (Eq. (5.5)) is scale-normalised hue stability; $\text{Chroma}_{\text{err}}$ (Eq. (5.6)) is the per-material chromaticity error against the pseudo-ground truth. Lower is better for all three. Red, orange, and yellow mark first, second, and third numerical rank; the bootstrap intervals and the paired tests of Table 5.3, not the colour fill, decide whether a difference is supported. The spread ratio (Eq. (5.7)) is a desaturation diagnostic and is not ranked. These are our own measures, so no published method reports comparable values; every row is evaluated locally under one pipeline.*

Model	$C_{\text{mat}} \downarrow$	$\text{Chroma}_{\text{err}} \downarrow$	$\text{Cast}_{\text{rel}} \downarrow$	Spread ratio (diag.)
Ours	0.157 [.141,.174]	0.129 [.109,.150]	0.425 [.360,.491]	0.999
CD-IID (Careaga and Aksoy, 2024)	0.190 [.172,.210]	0.090 [.073,.110]	0.306 [.262,.355]	0.944
RGB→X (Zeng et al., 2024)	0.128 [.115,.141]	0.203 [.170,.239]	0.338 [.273,.409]	0.907
Marigold-App (Ke et al., 2025)	0.193 [.173,.214]	0.195 [.152,.245]	0.355 [.309,.406]	0.728
Marigold-Light (Ke et al., 2025)	0.546 [.462,.645]	0.154 [.124,.187]	0.392 [.333,.459]	1.148
CRefNet (Luo et al., 2024)	0.151 [.138,.164]	0.201 [.165,.241]	0.355 [.309,.406]	0.484
Ordinal Shading (Careaga and Aksoy, 2023)	0.252 [.229,.279]	0.148 [.130,.166]	0.549 [.463,.637]	0.924

Table 5.3: *Paired differences on the held-out MID split, our model minus each external baseline, scene by scene. All three metrics are lower-is-better, so a negative entry means our model wins. Brackets are 95% percentile-bootstrap intervals over the 30 scenes (20,000 resamples); **bold** marks an interval clear of zero. Because every method is scored on the same scenes, the difference can be taken per scene, which cancels scene difficulty. Reproduce with `tests/eval/paired_bootstrap_mid.py`.*

Ours –	C_{mat}	$\text{Chroma}_{\text{err}}$	Cast_{rel}
CD-IID (Careaga and Aksoy, 2024)	-0.032 [-0.045,-0.019]	+0.039 [+0.022,+0.057]	+0.119 [+0.085,+0.156]
RGB→X (Zeng et al., 2024)	+0.030 [+0.014,+0.047]	-0.074 [-0.104,-0.045]	+0.087 [+0.024,+0.149]
CRefNet (Luo et al., 2024)	+0.007 [-0.004,+0.019]	-0.072 [-0.102,-0.044]	+0.070 [+0.030,+0.117]
Marigold-App (Ke et al., 2025)	-0.035 [-0.055,-0.016]	-0.067 [-0.103,-0.032]	+0.070 [+0.030,+0.111]
Marigold-Light (Ke et al., 2025)	-0.388 [-0.484,-0.308]	-0.025 [-0.055,+0.004]	+0.033 [-0.020,+0.085]
Ordinal Shading (Careaga and Aksoy, 2023)	-0.095 [-0.117,-0.075]	-0.019 [-0.035,-0.003]	-0.124 [-0.160,-0.094]

significantly less stable than RGB→X (+0.030 [+0.014, +0.047]). This comparison across methods mixes architectures, training data and model size, so it does not isolate the training strategy; the matched ablation of Sect. 5.6 does, and that is where the claim of this report rests.

Reading stability and accuracy together separates three behaviours among the seven rows. CRefNet is *stable but colourless*: a low Cast_{rel} next to a spread ratio of 0.484. RGB→X is *colourful but miscalibrated*: it keeps most of the spread (0.907) but on the wrong hues, with the worst $\text{Chroma}_{\text{err}}$ in the table (0.203). CD-IID is good on both. No single number captures all three, which is why we never report

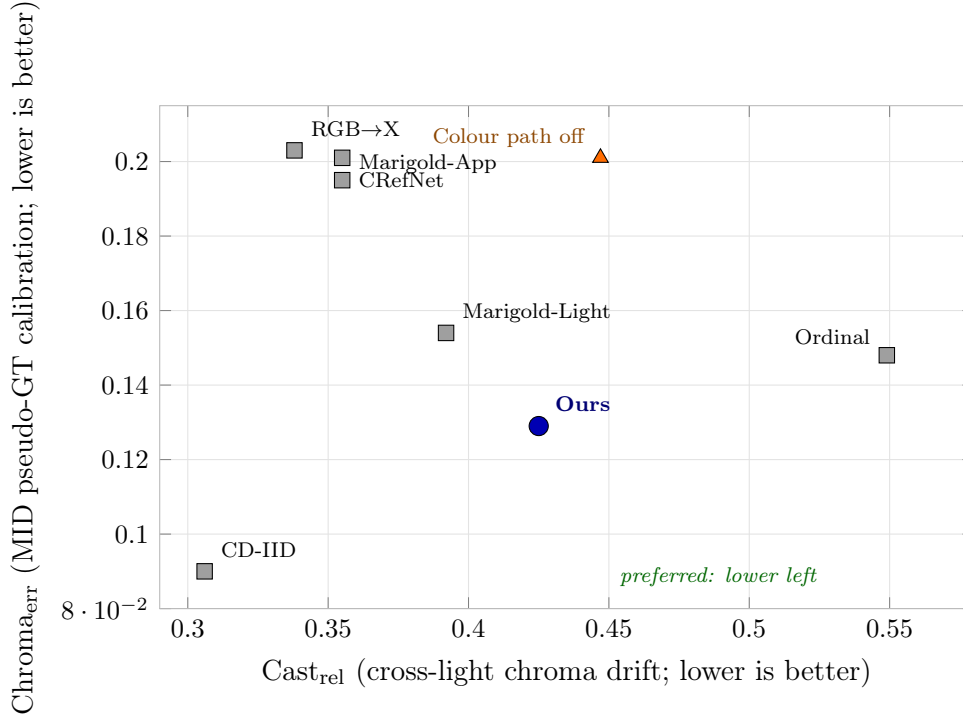


Figure 5.8: Hue drift across flash directions against chroma error to the pseudo-GT on the 30-scene MID test split; lower is better on both axes. CD-IID is best on both. Marigold-App and CRefNet drift less than our model but with substantially higher chroma error, and so does our own configuration without the colour path. Figure 5.9 shows whether low drift comes with lost colour.

stability without accuracy.

Our model trains 18.5M parameters against CD-IID’s 442.8M — every stage of which is optimised — and roughly 1290M for each diffusion baseline, at 0.148s per image against 0.254s and 0.8–1.0s (Table 5.1). The saving is in weights that had to be optimised, not in deployed size: the frozen backbone leaves us at 322.9M total, only $1.4\times$ below CD-IID.

Figure 5.9 shows the desaturation failure directly. Beside the pseudo-GT albedo, our own colour-path-off configuration and CRefNet are visibly drained of colour, while our model follows the reference more closely; these are also the two predictions that the pooled score of Eq. (5.2) would have ranked best.

Figure 5.8 plots hue stability against pseudo-GT chroma error for the roster. Lower is better on both axes. CRefNet and Marigold-App drift less than our model but with substantially higher chroma error, and the colour-path-off ablation sits in the same high-error region; CD-IID is best on both axes.

Figure 5.10 shows a qualitative example. Each column is a different flash

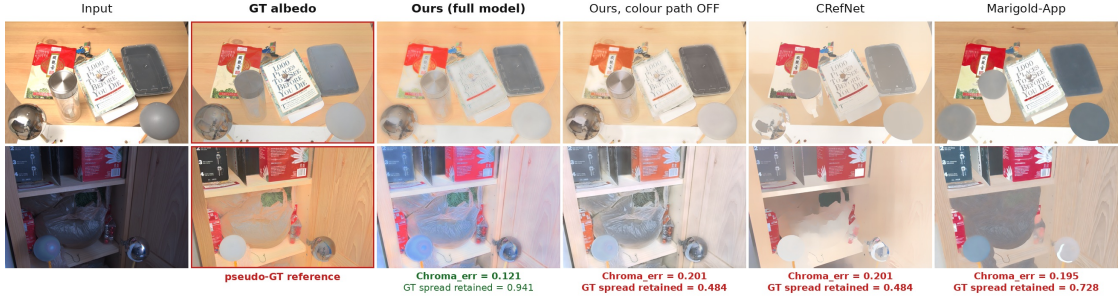


Figure 5.9: Chroma diagnostics on two distinct MIDIntrinsics test scenes and illuminant conditions (one row each). From left: the input, pseudo-GT albedo (red border), and four predictions. Albedo is defined only up to a global scale, so every panel is independently scale-normalised for display; no colour correction is applied, so the differences in saturation are differences in the predictions themselves. Values below each column are measured over the full 30-scene test split: $\text{Chroma}_{\text{err}}$ is the pseudo-GT calibration guard and the aggregate spread ratio diagnoses collapse. The colour-path-off row and CRefNet both retain 48.4% of pseudo-GT chroma spread and have error 0.201. The “Ours” column of this figure was rendered with a later checkpoint trained for 20k further steps with additional regularisers (printed values 0.121 and 94.1%); the reported model retains 99.9% with error 0.129. The figure will be regenerated with the reported model.



Figure 5.10: Qualitative albedo stability on one MID scene. The upper row shows four inputs and the pseudo-ground-truth albedo; the lower row shows the corresponding predictions and a per-pixel coefficient-of-variation map of predicted lightness (red = the prediction changes with the light). The map stays cool over the table and most of the matte packaging, and turns red on the plastic tray, the glass jar and the mirror sphere, surfaces an albedo-shading product cannot represent; the book cover also washes out under the brightest flash (light 12).

direction for the same scene, and the variation map shows where the predicted lightness changes with the light.

5.3 IIW: Perceptual Reflectance Ordering

IIW (Bell et al., 2014) has been the default benchmark of the field for a decade, so a decomposition method is expected to report it, and we do. It scores a prediction with the Weighted Human Disagreement Rate (WHDR): human annotators are shown two points in a photograph and asked which surface is lighter, and WHDR is the confidence-weighted fraction of those comparisons that the predicted albedo orders the wrong way,

$$\text{WHDR} = \frac{\sum_i w_i \cdot \mathbb{1}[\hat{r}_i \neq r_i]}{\sum_i w_i}, \quad (5.8)$$

where r_i is the human verdict on the i -th pair (lighter, darker, or equal), $\hat{r}_i$ is the verdict implied by the predicted albedo under a threshold on the ratio of the two points' intensities, and w_i is the annotator confidence. Lower is better.

Eq. (5.8) is worth writing out because its limitations are visible in it. The comparison $\hat{r}_i$ depends only on the *ratio* of two predicted intensities, so WHDR is invariant to any global rescaling of the albedo and, more importantly, it never looks at colour at all: the albedo is reduced to a lightness ordering before it is scored. A model can therefore tint every surface in the image with the colour of the lamp and lose nothing on this metric, as long as the relative lightness of surface pairs survives. Its labels are sparse and perceptual, resting on human judgements rather than a physical albedo measurement. Subjectivity is not inherently a weakness: when the intended property is perceived relative reflectance, people are the appropriate observers. The limitation is one of *scope*. A sparse ordinal lightness label cannot determine absolute intensity, chromaticity, or texture, so complementary metrics are needed for those properties (Wu et al., 2023; Sato et al., 2025).

This matters directly for the claim of this thesis. Coloured illuminant leakage is a *hue* error, and Eq. (5.8) is hue-blind by construction, so WHDR cannot confirm or refute the property we are trying to improve. We therefore report it as complementary evidence: it tests whether enforcing colour constancy damages the relative reflectance ordering that viewers care about, but it cannot validate the coloured-illumination claim by itself. The benchmark that does address material colour directly is MAW, which Sect. 5.4 turns to next and which was created for exactly this gap (Wu et al., 2023).

Table 5.4 separates locally evaluated rows from cited full-split context. Supervision status is stated explicitly because the local block contains both zero-shot models and models trained on IIW. The cited numbers are not used as a head-to-head comparison with our local rows: each source has its own inference implementation and training protocol.

Table 5.4: *Results on IIW (Bell et al., 2014). WHDR is lower-is-better. The upper block is evaluated locally; rows using IIW supervision are identified explicitly. The first cited block is selected no-IIW-supervision rows from Ordinal Shading (Careaga and Aksoy, 2023), Table 2. The second is the full-IIW roster reproduced by Wang et al. (Wang et al., 2025), Table I, an arXiv preprint that evaluates all 5,230 IIW images with per-method recommended settings. Repeated methods are intentional: they expose source- and protocol-dependent reported values, not a unified rerun. Both cited blocks are context rather than a head-to-head comparison with our local split. † marks a result obtained with IIW supervision under a different protocol.*

Model	WHDR ↓
Ours	0.286
Marigold-App (Ke et al., 2025)	0.193
Marigold-Light (Ke et al., 2025)	0.215
CRefNet (local; IIW-supervised) (Luo et al., 2024)	0.168
Ordinal (local) (Careaga and Aksoy, 2023)	0.257
<i>Cited from Ordinal Shading, Table 2 (no IIW supervision):</i>	
Bell et al. (Bell et al., 2014)	0.206
CGIntrinsics (Li and Snavely, 2018a)	0.178
NIID-Net (Luo et al., 2020)	0.166
PIE-Net (Das et al., 2022)	0.213
Ordinal Shading (Careaga and Aksoy, 2023)	0.249
<i>Full IIW test split, Wang et al., Table I:</i>	
Bell et al. (Bell et al., 2014)	0.206
CGIntrinsics (Li and Snavely, 2018a)	0.154
NIID-Net (Luo et al., 2020)	0.166
CRefNet (Luo et al., 2024)	0.145
Texture-aware IID (Wang et al., 2025)	0.136
<i>Original CRefNet result (different protocol):</i>	
CRefNet [†] (Luo et al., 2024)	0.108

Our model’s WHDR of 0.286 trails every locally evaluated baseline except Ordinal Shading. Within the ablation of Sect. 5.6, adding CIAI improves WHDR (0.334 to 0.283 without the colour path, 0.327 to 0.286 with it), so the pair losses help perceived reflectance ordering rather than hurt it; the gap to other methods lies in the base model.

Figure 5.11 shows qualitative results on the IIW test set, and illustrates the mismatch between what the metric rewards and what this thesis is optimising.



Figure 5.11: Qualitative albedo predictions from our model on IIW test images. Each row shows an input image (left) and its predicted albedo (right). The predictions preserve broad material regions but show the two competing errors that an ordinal score can expose only at annotated point pairs: legitimate material boundaries may be over-smoothed, while illumination structure remains on the ceiling and around the windows. This figure is an honest failure case rather than evidence of colour accuracy, because IIW provides no measured RGB albedo.

5.4 MAW: Measured Real-World Albedo

MAW (Wu et al., 2023) was built to close the gap that Sect. 5.3 just described. Instead of asking people which of two patches looks lighter, its authors recover a physical albedo value for real surfaces. They photograph a surface twice, once with a grey card of known reflectance resting on it and once without. Both the surface and the card receive the same illumination, so the shading term is shared and cancels in the ratio of the two measurements, which leaves the surface’s albedo in absolute terms. The ground truth is therefore a measured quantity rather than a perceptual vote, which makes MAW the benchmark best able to answer the question this thesis poses: when a real room is lit by a coloured light, does the predicted material colour match the material’s real colour?

Because that ground truth is measured rather than derived from the prediction, MAW scores colour *correctness* against an external reference, not merely stability across conditions. Both metrics remove the global-scale ambiguity: ΔE is computed on chromaticity, while the intensity error is explicitly scale-invariant. MAW therefore measures the right *chromaticity* and the right *relative* brightness, not an absolute radiometric value. It is thus complementary to MID rather than a substitute for it: a model can be perfectly stable across illuminants (a low C_{mat}) and still be consistently wrong about what colour the wall is, and MAW is what catches that.

The public scorer first resizes the prediction and region mask to 320×240

and extracts a robust median RGB vector $\hat{\mathbf{a}}_m$ for each measured material region m ; $\mathbf{a}_m^*$ is the corresponding measured reference and w_m is the region area. Let $\ell(\mathbf{x}) = \frac{1}{3} \sum_c x_c$ denote the scorer’s RGB channel mean. The colour score removes intensity independently for each material, $\beta_m = \ell(\mathbf{a}_m^*)/\ell(\hat{\mathbf{a}}_m)$, and reports the area-weighted CIEDE2000 colour difference,

$$\Delta E_{00} = \frac{\sum_m w_m \Delta E_{00}(\text{Lab}(\beta_m \hat{\mathbf{a}}_m), \text{Lab}(\mathbf{a}_m^*))}{\sum_m w_m}. \quad (5.9)$$

This per-material scale alignment makes the score primarily a chromaticity test rather than a test of recovered intensity. It is therefore the primary external score for a coloured-illuminant leak.

The second score tests relative material brightness. Writing $\hat{l}_m = \ell(\hat{\mathbf{a}}_m)$ and $l_m^* = \ell(\mathbf{a}_m^*)$, the scorer fits one area-weighted global scale

$$\alpha^* = \frac{\sum_m w_m \hat{l}_m l_m^*}{\sum_m w_m \hat{l}_m^2}, \quad \text{SI-MSE} = \frac{\sum_m w_m (\alpha^* \hat{l}_m - l_m^*)^2}{\sum_m w_m}. \quad (5.10)$$

Because the scale is fitted away, this score is insensitive to a uniformly too-dark or too-bright prediction and instead penalises the wrong *relative* lightness of different materials. It does not directly measure spatial blur, although smoothing can change the robust region medians on which it is computed.

We run every local method with the common 1280-pixel long-side inference cap specified in Sect. 5.1; the public scorer then performs its fixed 320×240 evaluation and scale handling, matching the MAW protocol used by CD-IID (Careaga and Aksoy, 2024).

Table 5.5 reports ΔE , the CIELAB colour difference between the prediction and measured albedo, and intensity SI-MSE $\times 100$, which removes the global-scale ambiguity before assessing brightness. Lower is better for both. The cited block is retained only for methods reported under the same MAW protocol, with CD-IID Table 1 as the primary external reference.

On measured albedo our model is mid-table: its colour error ($\Delta E = 4.155$) is behind CD-IID, Marigold-App and CRefNet, and its relative-intensity error (0.504) is among the lowest. CIAI itself slightly worsens both MAW measures in the ablation (4.115 to 4.155 and 0.395 to 0.504 with the colour path), so lightness stability on MID is not bought with better single-image accuracy; the stability gain and the accuracy cost are reported together.

Figure 5.12 shows a representative MAW scene. The ground-truth panel is masked to the surface regions whose albedo was physically measured by the dataset authors; only those regions enter the score.

Table 5.5: Results on MAW (Wu et al., 2023), evaluated with the benchmark scorer. ΔE measures CIELAB colour difference and intensity SI-MSE $\times 100$ measures scale-invariant brightness error; lower is better for both. The upper block is evaluated locally. The complete lower block is transcribed from CD-IID (Careaga and Aksoy, 2024), Table 1. The colour-path-only row is Row 3 of the ablation (no CIAI), shown for reference. * denotes the grayscale-shading assumption, for which predicted chromaticity is inherited from the input image. Red, orange, and yellow cells mark first, second, and third place over the common protocol.

Model	$\Delta E \downarrow$	Int. SI-MSE $\times 100 \downarrow$
Ours	4.155	0.504
Ours, colour path only (ablation Row 3)	4.115	0.395
Marigold-App (Ke et al., 2025)	<u>3.775</u>	0.653
Marigold-Light (Ke et al., 2025)	4.214	<u>0.493</u>
CRefNet (Luo et al., 2024)	<u>3.970</u>	0.942
Ordinal (local) (Careaga and Aksoy, 2023)	6.884	0.657
<i>Cited from CD-IID, Table 1:</i>		
Bell et al. (Bell et al., 2014)	6.61	3.11
CGIntrinsics (Li and Snavely, 2018a)	5.15	2.71
Li & Snavely (2018a)* (Li and Snavely, 2018b)	6.56	1.72
NIID-Net (Luo et al., 2020)	4.73	1.24
IRISformer (Zhu et al., 2022b)	4.94	1.44
Kocsis et al. (Kocsis et al., 2024)	5.35	1.13
IntrinsicAnything (Chen et al., 2024)	4.12	0.98
Ordinal Shading* (Careaga and Aksoy, 2023)	6.56	0.57
CD-IID (Careaga and Aksoy, 2024)	3.37	0.54

5.5 ARAP: Constancy and Albedo Accuracy

Status of these numbers. The ARAP results in this section are *provisional*. After they were computed we found that ARAP stores its ground-truth albedo in three incompatible encodings: 40 scenes divided by the Radiance constant 179 (maximum about 0.0056), seven 8-bit sRGB scenes, and four unscaled ones. Our evaluator masks pixels with a fixed absolute threshold (albedo luminance > 0.004), which keeps a median of 1.5% of the frame for the first group, about 99% for the others, and leaves six scenes with an empty mask; this is also why only 136 of 157 images are scored. Every model passed through the same mask, so the comparisons are internally consistent, but the mask favours bright regions and no absolute ARAP value here should be compared with published numbers. In a check that



Figure 5.12: Qualitative comparison on three content-diverse MAW photographs (one per row): a kitchen, a dark office, and a living area. Left to right: input photograph, measured albedo (red border and masked to reliable regions), and the prediction from Ours. All three middle panels show the corresponding physically measured MAW albedo mask.

changed only the mask, re-scoring one of our checkpoints with an encoding-aware mask raised its LMSE by 79% and lowered its SSIM by 40%, which shows how much is at stake. The tables will be regenerated with `tests/eval/arap_preprocess.py`; until then no claim in this report rests on ARAP alone.

ARAP (Bonneel et al., 2017) is synthetic, and we report it last for that reason: a rendered scene is an easier target than a photograph, so a good ARAP number cannot by itself support a claim about real rooms. What ARAP uniquely offers is that it is the only benchmark here with dense ground-truth albedo *and* several renderings of the same scene under different lights. MID has the repeated illuminants but no true albedo; MAW has the true albedo but only one illuminant per surface. ARAP has both, so it is the one place where constancy and accuracy can be measured on the very same pixels, and where a claim that improving one damages the other can actually be tested.

It answers two distinct questions, which must not be merged. First, does a prediction stay stable when the original coloured illumination changes? This is

tested on the raw renderings with our self-consistency measures (Table 5.6). Second, how accurate is the recovered albedo when the input is neutrally lit? This is tested on white-balanced input against dense ground-truth albedo (Table 5.7).

We white-balance correctly, which is worth stating because an earlier version of our evaluator did not. A single global exposure rescale (multiplying all three channels by one scalar) leaves the chromaticity of the illuminant untouched and is therefore *not* white balance; measured on a coloured-lit ARAP scene it leaves the implied illuminant as far from neutral after the operation as before. We instead reconstruct an achromatic-illuminant image from the ground-truth albedo: writing the implied shading as $\mathbf{S} = \mathbf{I}/\mathbf{A}^*$, we recompose $\mathbf{I}_{\text{wb}} = \mathbf{A}^* \cdot \text{lum}(\mathbf{S})$, which drives the implied illuminant chromaticity to exactly neutral. The accuracy table uses this reconstruction; the constancy table uses the raw coloured renderings, since removing the illuminant would erase the very variable the constancy measure exists to probe.

Two coverage points follow. The constancy table (Table 5.6) is restricted to the multi-illuminant groups whose illuminant *colour* actually changes between renderings (16 groups overall, 11 indoor): cross-light drift is undefined for a single-light scene, and groups that vary only intensity or direction carry no colour cast for an invariant albedo to be tested against, so including them would pull every model’s score toward zero and mask the effect the thesis is about. The accuracy table (Table 5.7) instead scores every image with usable dense albedo, namely 136 of the 157 in the collection; the other 21 are left with fewer than 88 valid pixels by the evaluation mask described above and are dropped for every model alike. The accuracy table combines our local white-balanced evaluation with published original-ARAP results from Ordinal Shading (Careaga and Aksoy, 2023), Table 1.

Figure 5.13 provides the model-by-illuminant comparison that the aggregate tables cannot show spatially. Figure 5.14 then samples three distinct ARAP interiors to reduce dependence on that single scene.

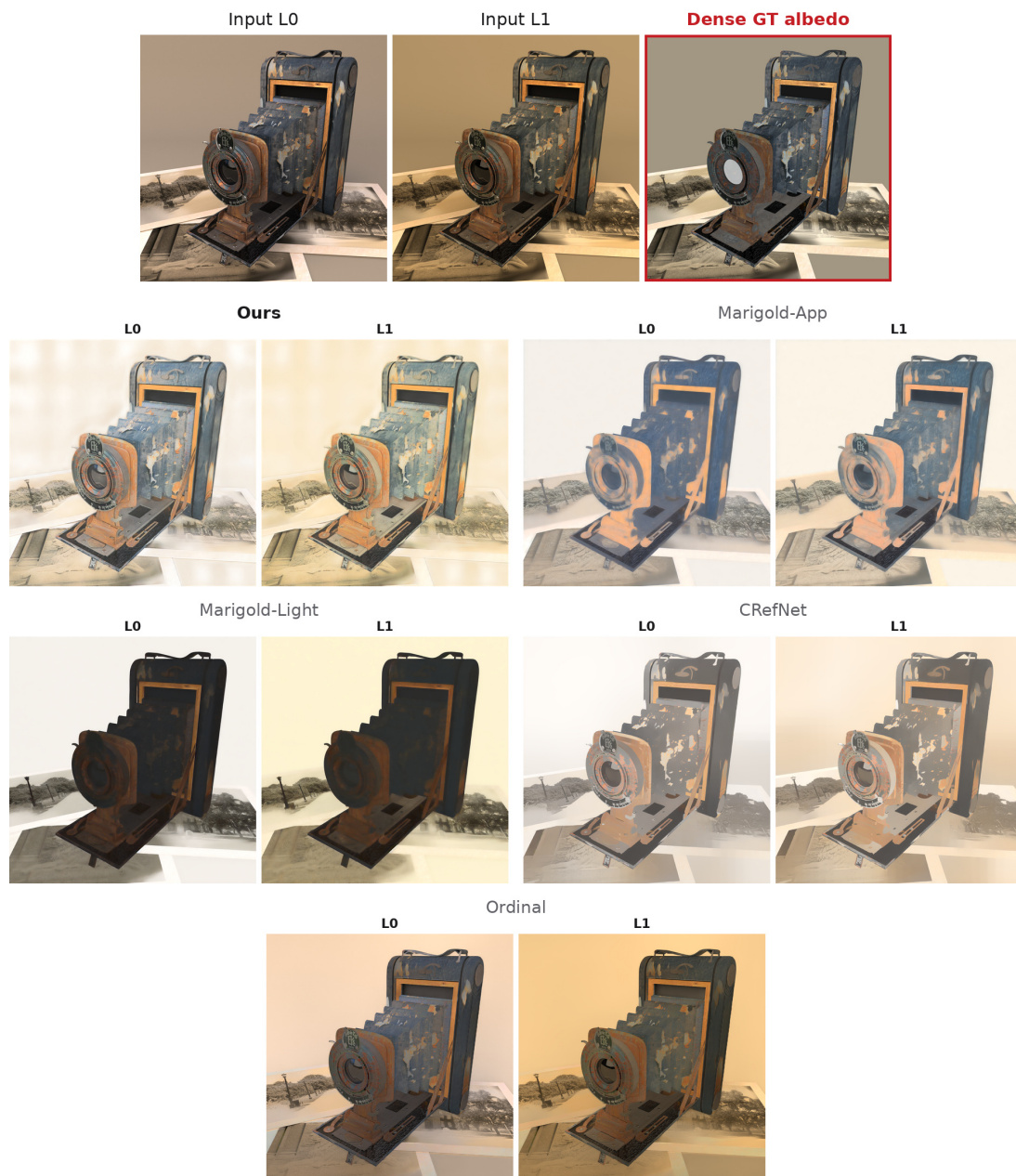


Figure 5.13: Model-by-illuminant comparison on the ARAP camera scene. The reference strip shows both rendered illuminant conditions and the dense ground-truth albedo once. The grid below shows each locally evaluated model’s albedo under the same two illuminants, so the reader can compare consistency without repeatedly duplicating the reference. Predictions are independently scale-normalised for display, with no colour correction. The row labelled *Ours* uses the model selected for the thesis qualitative figures. The grid exposes different failure modes rather than declaring a winner: *CRefNet* suppresses much of the material colour, *Marigold-Light* and *Ordinal* retain illumination structure, and our *L1* output still contains hard-shadow leakage. *Marigold-App* is comparatively stable on this scene; the full-set metrics in Table 5.6 and Table 5.7 remain the aggregate evidence.

Table 5.6: Provisional; see the status note at the start of Sect. 5.5. ARAP raw-input constancy on the colour-varying groups of the original ARAP collection (Bonneel et al., 2017) — the groups in which the illuminant colour changes between renderings, which is the literal case this thesis is about (16 groups overall, 11 indoor; the remaining multi-light groups vary only intensity/direction and are excluded, since they carry no colour for the albedo to be invariant to and would dilute the score toward zero). C_{arap} measures cross-illumination albedo variation; lower is better. Red, orange, and yellow mark first, second, and third numerical rank in each column; displayed ties share a rank. These are internal diagnostics, evaluated locally.

Model	All ($n=16$)	Indoor ($n=11$)
	$C_{\text{arap}} \downarrow$	$C_{\text{arap}} \downarrow$
Ours	0.154	0.119
Marigold-App (Ke et al., 2025)	0.113	0.113
Marigold-Light (Ke et al., 2025)	0.340	0.340
CRefNet (Luo et al., 2024)	0.128	0.129
Ordinal Shading (Careaga and Aksoy, 2023)	0.175	0.165

The pooled Cast_{RMS} of Eq. (5.2) is deliberately not reported here: Sect. 5.2 shows that an absolute chroma variance is minimised just as effectively by a colourless albedo as by an invariant one, so it cannot stand as a constancy score. C_{arap} is likewise a pure constancy score and can be minimised by a constant prediction; it must be read beside the dense-GT accuracy guard in

Table 5.7.

Table 5.7: Provisional; see the status note at the start of Sect. 5.5. ARAP albedo accuracy, within-study, on the full Bonneel et al. collection (Bonneel et al., 2017) (136 of 157 images have dense ground-truth albedo with enough valid pixels to score; the same 136 are used for every model). Input is white-balanced by reconstructing an achromatic-illuminant image from the ground-truth albedo (Sect. 5.5). LMSE and mean-normalised RMSE are lower-is-better, SSIM higher. The upper block is evaluated locally under one harness; the lower block is transcribed from Ordinal Shading (Careaga and Aksoy, 2023), Table 1, and reports no SSIM because that protocol is unspecified (see the note below the table). Red, orange, and yellow mark first, second, and third place; LMSE and mn-RMSE rank across all rows, SSIM within the local block.

Model	LMSE ↓	mn-RMSE ↓	SSIM ↑
Ours	0.0279	0.2259	0.611
Marigold-App (Ke et al., 2025)	0.0266	0.2529	0.607
Marigold-Light (Ke et al., 2025)	0.0706	0.6187	0.589
CRefNet (Luo et al., 2024)	0.0223	0.2916	0.622
<i>Published original-ARAP protocol (Ordinal Shading, Table 1):</i>			
Chromaticity (Bonneel et al., 2017)	0.024	0.296	—
NIID-Net (Luo et al., 2020)	0.022	0.206	—
Li & Snavely (Li and Snavely, 2018b)	0.018	0.227	—
Bell et al. (Bell et al., 2014)	0.028	0.317	—
PIE-Net (Das et al., 2022)	0.026	0.273	—
Ordinal Shading (Careaga and Aksoy, 2023)	0.021	0.252	—

Why no published SSIM. We transcribe a published metric only when its protocol is specified well enough for the number to mean the same thing as ours. That test fails for SSIM: Ordinal Shading’s supplement fixes only the invalid-pixel threshold (albedo < 0.004, §A.6/B.1) and lists SSIM among its “scale-invariant metrics” without stating the alignment, and the shared `chrislib` benchmark library implements LMSE, si-RMSE and WHDR but no SSIM at all. Running Ordinal Shading itself through our closest reconstruction yields 0.671 against the 0.761 it reports, so the gap is a protocol difference we cannot resolve rather than a quality difference. The local SSIM column is still internally valid: all five rows share one harness.

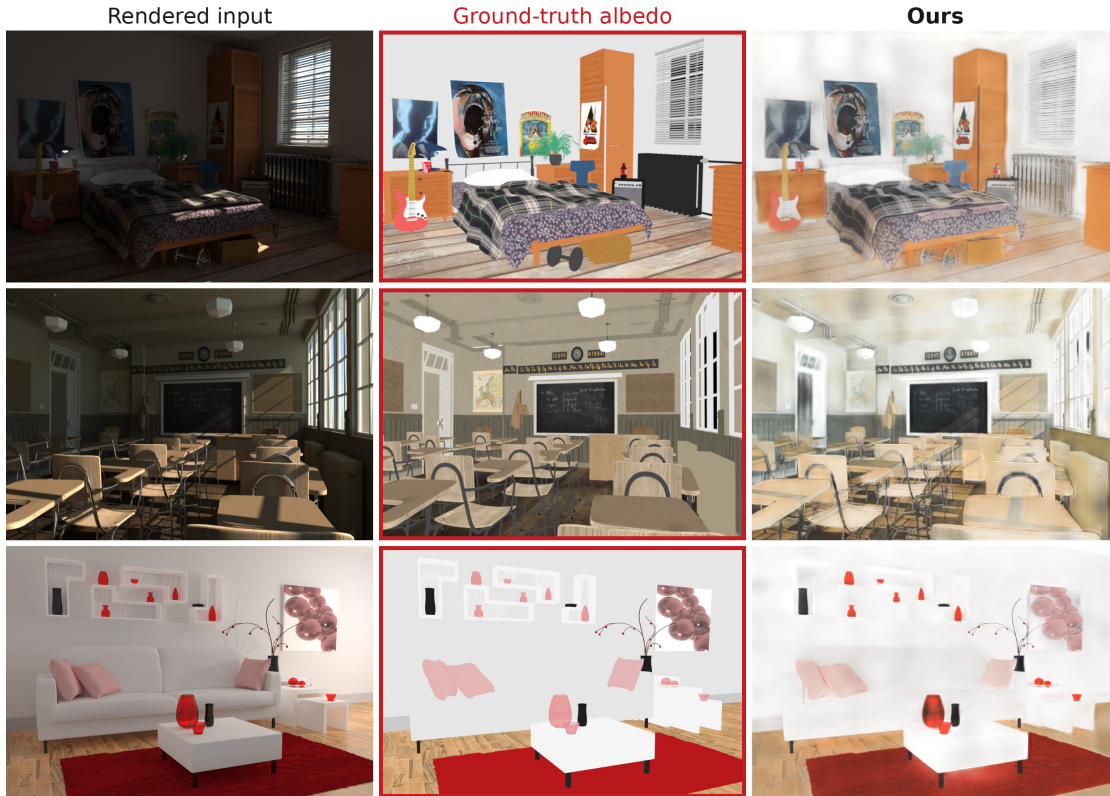


Figure 5.14: Qualitative accuracy on three content-diverse ARAP scenes: bedroom, classroom, and livingroom. Each row shows the rendered input, dense ground-truth albedo (red border), and the prediction from Ours. This figure tests scene diversity and dense-albedo fidelity; Figure 5.13 separately compares models while the illuminant changes.

5.6 Ablation Study

The benchmark sections placed our model against external methods, which differ from it in architecture, data and size at once. This section isolates what CIAI itself contributes, with a 2×2 ablation inside our own architecture.

The ablation isolates the contribution of CIAI by comparing configurations along two axes: whether the cross-illumination losses ($\mathcal{L}_{\text{inv}}$, $\mathcal{L}_{\text{expl}}$) are active, and whether the colour-path bundle (wide RGB skip and its chroma-loss setting) is active (Sect. 3.2). These are the two components that, according to the coupling argument in that section, are expected to interact. Table 5.8 summarises the resulting 2×2 design. Part 1 reports self-consistency on raw-coloured inputs: the MID measures with their fidelity guard, then ARAP constancy over all eligible groups and over the domain-matched indoor subset. Part 2a reports measured-albedo accuracy, the GT-achromatized ARAP accuracy guard, and ancillary IIW ordering. Part 2b reports raw-input ARAP accuracy over the same all-ARAP and indoor subsets as Part 1. The GT-achromatized ARAP columns support the within-study comparison but are not presented as a reproduced published score. All four configurations share the same 19k initialization, data curriculum, and 40k training horizon. The two within-architecture comparisons are the causal CIAI tests: Row 1→Row 2 toggles cross-illumination losses with the colour path off, and Row 3→Row 4 does the same with the colour path on. These two within-architecture contrasts are the matched CIAI tests. The cross-axis colour-path comparison is not causal: it changes head parameters and chroma supervision, starts Adam afresh, and the recorded fallback produces an approximately $1.85\times$ different effective learning-rate trajectory. We therefore report cross-axis differences descriptively rather than attributing them to the skip alone.

All rows are single training runs, so small differences should not be read as effects.

Effect of cross-illumination losses within each architecture. The clean CIAI comparisons are Row 1→Row 2 and Row 3→Row 4, the two contrasts in which only the cross-illumination losses change. The clear effect is on *lightness* stability: cross-illumination supervision lowers C_{mat} from 0.250 to 0.180 with the colour path off and from 0.259 to 0.157 with it on, corresponding to a 28% and 39% reduction, both well outside the scene-bootstrap intervals of Table 5.2. On scale-normalised hue invariance the effect is present but small: Cast_{rel} moves from 0.477 to 0.447 and from 0.444 to 0.425. The relevant paired bootstrap, rather than overlap of marginal intervals, gives Row 1-minus-Row 2 0.030 with a 95% interval [0.011, 0.049], and Row 3-minus-Row 4 0.019 with interval [0.007, 0.031]. Both are modest but significant hue-stability gains, but in relative terms (-6%

Table 5.8: CIAI study across cross-illumination losses and the colour-path bundle. All configurations share the same 19k initialization and data curriculum, and 40k horizon; Row 1→Row 2 and Row 3→Row 4 are the matched cross-illumination comparisons. Cross-axis comparisons are descriptive because the colour path also changes chroma supervision, optimizer state, and effective learning-rate history. Part 1 reports raw-input constancy on the colour-varying ARAP groups (the same 16/11 subset used in Table 5.6); all ARAP columns are provisional (Sect. 5.5); Part 2a measured-albedo accuracy, the GT-achromatized ARAP guard and ancillary IIW ordering; Part 2b raw-input ARAP accuracy. Lower is better everywhere except ARAP SSIM, and the aggregate spread ratio, whose target is 1.000. Best per column is **bold**. Figure 5.15 shows the corresponding albedo and variation maps.

Part 1: Raw-input constancy (self-consistency across illuminants)

Row	Cross-illumination	Colour path	MID (30 scenes)				ARAP all	ARAP indoor
			$C_{\text{mat}} \downarrow$	$\text{Cast}_{\text{rel}} \downarrow$	$\text{Chroma}_{\text{err}} \downarrow$	Spread ratio ($\rightarrow 1$)	$C_{\text{arap}} \downarrow$	$C_{\text{arap}} \downarrow$
1	off	off	0.250	0.477	0.186	0.512	0.164	0.122
2	on	off	0.180	0.447	0.201	0.484	0.145	0.108
3	off	on	0.259	0.444	0.124	0.925	0.155	0.117
4	on	on	0.157	0.425	0.129	0.999	0.154	0.119

MID spread ratios use the ratio of aggregate predicted and pseudo-GT spreads, not the biased mean of per-scene ratios. The pooled ARAP Cast_{RMS} columns are omitted for the reason given under Table 5.6: an absolute chroma variance is minimised as effectively by a colourless albedo as by an invariant one. The MID fidelity metrics cannot replace them on ARAP, which ships no material segmentation, so ARAP contributes constancy (C_{arap}) and the dense-GT accuracy of

Part 2a instead.

Part 2a: Standard-protocol accuracy and structure (against ground truth)

Row	Cross-illumination	Colour path	MAW		ARAP (GT-achromatized)			IIW
			$\Delta E \downarrow$	$\text{Int.} \times 100 \downarrow$	$\text{LMSE}^\dagger \downarrow$	$\text{mn-RMSE}^\dagger \downarrow$	$\text{SSIM}^\dagger \uparrow$	$\text{WHDR} \downarrow$
1	off	off	4.434	0.409	0.0307	0.2378	0.602	0.334
2	on	off	4.631	0.474	0.0320	0.2313	0.589	0.283
3	off	on	4.115	0.395	0.0283	0.2348	0.617	0.327
4	on	on	4.155	0.504	0.0279	0.2259	0.611	0.286

Part 2b: Raw-input ARAP albedo accuracy (against ground truth)

Row	Cross-illumination	Colour path	All ARAP (raw input)		Indoor ARAP (raw input)	
			$\text{LMSE}^\ddagger \downarrow$	$\text{mn-RMSE}^\ddagger \downarrow$	$\text{LMSE}^\ddagger \downarrow$	$\text{mn-RMSE}^\ddagger \downarrow$
1	off	off	0.0463	0.341	0.0441	0.323
2	on	off	0.0397	0.292	0.0370	0.275
3	off	on	0.0426	0.319	0.0400	0.301
4	on	on	0.0396	0.293	0.0382	0.276

$\dagger$ GT-achromatized ARAP accuracy guard (136 of 157 images): an internal within-study comparison, not a reproduced literature table. IIW WHDR is ancillary and is not a selection criterion. $\ddagger$ Raw-input ARAP accuracy is a local multi-illuminant stress test and is not compared with published white-balanced results.

and -4%) five to nine times smaller than the lightness gain (-28% and -39%). That asymmetry is the first sign of the limit examined in Sect. 5.7.

The pseudo-GT calibration guard supplies the necessary qualification. With the colour path off, $\text{Chroma}_{\text{err}}$ worsens from 0.186 to 0.201; with it on, the change from 0.124 to 0.129 is not significant. CIAI therefore improves paired constancy but does not itself improve colour calibration. It also improves ancillary IIW WHDR in the first pair, while both MAW measures worsen slightly. The load-bearing result is the repeated paired constancy gain, read beside rather than instead of the external accuracy guards.

The colour path. The colour path is a design choice of the base model, not part of CIAI. The rows without it keep about half of the pseudo-GT chroma spread whether CIAI is off (Row 1, 0.512) or on (Row 2, 0.484); the rows with it keep 0.925 and 0.999, and have lower $\text{Chroma}_{\text{err}}$ and better MAW colour, Row 3 obtaining the best ΔE (4.115) and intensity score (0.395). This is a descriptive comparison: the bundle changes chroma supervision, head parameters, optimiser state and effective learning-rate history at once, so it does not isolate the skip from the chroma loss. Row 4 is the most stable of the four on both lightness and hue, measured as drift relative to the model’s own chroma (Eq. (5.5)).

CIAI also makes the albedo more accurate on raw input. The pair losses are not only a stability constraint. On ARAP renderings fed with their original coloured illumination (Part 2b), adding CIAI lowers the dense albedo error in both matched pairs: LMSE from 0.0463 to 0.0397 (-14%) and mean-normalised RMSE from 0.341 to 0.292 (-14%) without the colour path, and LMSE from 0.0426 to 0.0396 (-7%) and RMSE from 0.319 to 0.293 (-8%) with it; the indoor subset moves the same way ($-16\%/ -15\%$ and $-5\%/ -8\%$). On the same scenes white-balanced against the ground truth (Part 2a) the change is small and mixed (LMSE $+4\%$ and -1% , RMSE -3% and -4%). The gain therefore appears where the input still carries its lighting, which is what training on raw pairs should teach, and it appears on a synthetic benchmark the model never trained on. These ARAP numbers share the provisional evaluation mask of Sect. 5.5; because all four rows pass through the same mask, the matched differences are more reliable than the absolute values, but they should be confirmed once ARAP is regenerated.

Why Row 4 is our model. Row 4 is the complete CIAI configuration: the most stable of the four rows on both MID measures, the full pseudo-GT chroma spread (ratio 0.999), and the lowest raw-input ARAP error with the colour path. Its weaknesses, a slightly higher $\text{Chroma}_{\text{err}}$ than Row 3 (0.129 against 0.124) and



Figure 5.15: Ablation: the four CIAI configurations on one MID scene under three illuminant conditions ($L0$, $L10$, $L20$). Columns: the three input conditions, the pseudo-ground-truth albedo (red box), the three predicted albedos, and two cross-condition variation heatmaps computed across five illuminant conditions ($L0$, $L5$, $\dots$, $L20$; 5, three of which are shown): luminance CoV, matching MID C_{mat} , and the legacy pooled chroma-cast CoV diagnostic $Cast_{RMS}$ ($\sqrt{\text{Var}(R/G) + \text{Var}(B/G)}$ across conditions). Rows, top to bottom: no CIAI (Row 1), cross-illumination only (Row 2), colour-skip only (Row 3), and full CIAI (Row 4). These are local, single-scene illustrations rather than estimates of the full-split metrics in Table 5.8: the heatmaps expose where the predicted albedo changes with illumination, while the table establishes the dataset-level ranking. In particular, chroma variation is spatially and scene dependent, so the figure is not used to infer the global $Cast_{RMS}$ ordering.

a worse MAW score (4.155 against 4.115 ΔE , 0.504 against 0.395 intensity), are reported in Table 5.8 and in Sect. 5.4.

Figure 5.15 shows the qualitative effect of each ablation row.

5.7 Colour Stability: A Limitation

The results so far share a pattern: CIAI improves lightness stability substantially and hue stability barely. This is expected for two reasons. First, $\mathcal{L}_{\text{expl}}$ uses luminance only (Sect. 3.3), so nothing in CIAI states where a change in illuminant colour must go. Second, MID’s illumination changes little in colour. This section measures the second point and checks whether our hue drift follows the illuminant colour.

5.7.1 How much colour does MID’s illumination vary?

We measure the illuminant of each frame on its grey probe and take the distance between the chromaticities (R/Σ , B/Σ) of two frames of a scene as the *pair gap*.

Table 5.9: Median chromaticity gap between the illuminants of two frames of one scene. MID is measured over all frame pairs of the 30 test scenes; ARAP over all measurable pairs.

Corpus	Median gap	vs. MID
MID (training pairs and stability benchmark)	0.030	1.0×
ARAP, all measurable pairs	0.088	2.9×
ARAP, indoor pairs	0.070	2.3×

Table 5.10: Mean $Cast_{rel}$ per tercile of scene illuminant gap on the MID test split, and the ratio of the high to the low tercile. Re-aggregated from the per-scene results behind Table 5.2; no new inference.

Model	Shading	Low	Mid	High	High / low
Ours	RGB	0.368	0.379	0.528	1.43
CRefNet	grey	0.313	0.310	0.443	1.42
Ordinal Shading	grey	0.453	0.489	0.705	1.56
Marigold-Light	RGB	0.370	0.384	0.422	1.14
Marigold-App	RGB	0.368	0.316	0.381	1.03
CD-IID	grey + chroma stage	0.309	0.295	0.315	1.02
RGB→X	RGB	0.361	0.328	0.325	0.90

For ARAP we read the illuminant from $\mathbf{I}/\mathbf{A}^*$ instead. Table 5.9 compares the two.

86% of MID pairs fall below 0.08, and per-scene medians span only 0.017 to 0.086. MID’s flash is white; what colour change there is comes from light bounced off coloured surfaces. MID therefore teaches lightness well and colour weakly, and the hue stability measured on it is measured over a narrow range.

5.7.2 Does our hue drift follow the illuminant colour?

If a model’s hue drift is caused by the illuminant colour, it should be larger on scenes whose illuminant colour changes more. We split the 30 test scenes into terciles by their median pair gap and divide the mean $Cast_{rel}$ of the top tercile by that of the bottom one; a ratio of 1 means the drift does not follow illuminant colour.

Table 5.10 gives two readings. Our model’s hue drift rises by 43% on the scenes where the illuminant colour moves most, as much as CRefNet’s, whose grey shading cannot represent a coloured light at all; CIAI’s small hue gain in the ablation (Sect. 5.6) has not removed this dependence. And CD-IID, the one method with a dedicated shading-chroma stage, shows no dependence on illuminant colour (1.02)

while keeping its albedo colour (spread ratio 0.944). MID’s narrow colour range makes this a low-powered test, but the contrast is large and consistent.

Table 5.10 shows that our model’s hue drift rises by 43% on the scenes where the illuminant colour moves most, as much as CRefNet’s, whose grey shading cannot represent a coloured light at all; CIAI’s small hue gain in the ablation (Sect. 5.6) has not removed this dependence. CD-IID, the one method with a dedicated shading-chroma stage, shows no dependence on illuminant colour (1.02) while keeping its albedo colour (spread ratio 0.944). MID’s narrow colour range makes this a low-powered test, but it agrees with the ablation. Colour stability therefore needs a model that predicts the shading colour separately, which is the plan of Chapter 7.

6 | In Practice

The previous chapter measured the decomposition; this chapter puts it to use. Separating a photograph into an albedo and a three-channel shading layer is not only an estimation target but a practical handle: once the two factors are available, an image can be altered by changing one of them and leaving the other untouched, which no edit on the entangled pixels can do. This chapter develops that idea into a single application, *factor-space editing*, and presents two instances of it that follow directly from the properties established in Chapter 5.

This chapter is a proof of concept, not a second evaluation. Each edit is defined directly on the predicted factors, and its behaviour follows from how those factors are constructed, so the argument is made by formulation rather than by a quantitative application study, for which no user study or application-specific metric is available. Every factor comes from the single trained model selected in Sect. 5.5, labelled simply *Ours*; the chapter draws no comparison between internal checkpoints. That model’s decomposition is shown first (Figure 6.1), and the two edits are then read as operations on its predicted albedo, shading, and residual.

All edits below start from the decomposition $\mathbf{I} = \mathbf{A}_d \odot \mathbf{S}_d + \mathbf{R}$, where $\mathbf{A}_d$ is the predicted diffuse albedo, $\mathbf{S}_d$ the coloured diffuse shading, and $\mathbf{R} = (\mathbf{I} - \mathbf{A}_d \odot \mathbf{S}_d)_+$ the analytic non-diffuse residual that captures specular highlights, visible light sources, and clipped regions (Sect. 3.1).



Figure 6.1: Predicted decomposition by Ours on a realistic indoor photograph. From left: input photograph; predicted diffuse albedo $\mathbf{A}_d$; diffuse shading $\mathbf{S}_d = (1 - \boldsymbol{\pi})/\boldsymbol{\pi}$ obtained by inverting the model’s inverse-domain shading output $\boldsymbol{\pi}$; and the analytic residual $\mathbf{R} = (\mathbf{I} - \mathbf{A}_d \odot \mathbf{S}_d)_+$. Albedo and shading are independently normalised for display. The residual is shown at its calculated magnitude, without amplification, so it remains dark except at bright non-diffuse regions such as glass and highlights. This is a qualitative factorisation, not ground truth. The four panels are direct crops from the saved visualization; no new inference or post-hoc colour correction is applied for this figure.

We do not include a cross-illuminant transfer demonstration, in which one scene is re-rendered under another scene’s illuminant. Its natural score, the PSNR of the recomposed image, is highest for the visibly desaturated colour-path-off ablation, because compensating albedo and shading errors cancel in their product; it is a gameable factor-consistency diagnostic rather than evidence of physical correctness. The two edits below instead each act on a single factor of the decomposition, one on the residual $\mathbf{R}$ and one on the shading $\mathbf{S}_d$, so their behaviour follows directly from how those factors are defined.

6.1 Factor-Space Editing

Residual-space suppression. The residual layer contains image energy not represented by the predicted diffuse product. This can include specular highlights and clipped sources, but it can also include ordinary scene structure that the estimated albedo and shading fail to reconstruct. Scaling it yields the edit

$$\mathbf{I}'_{\alpha} = \mathbf{A}_d \odot \mathbf{S}_d + \alpha \mathbf{R}, \quad \alpha \in [0, 1]. \quad (6.1)$$

Setting $\alpha < 1$ dims energy assigned to the residual while holding the model’s predicted diffuse product fixed; $\alpha = 0$ removes that component entirely. The same

operation performed in image space (for example, lowering the exposure of bright regions) also changes diffuse regions; residual scaling instead holds the model’s predicted diffuse product fixed. This is the same operation used for highlight editing by Careaga and Aksoy (2024), but whether it behaves as a selective highlight control depends on the quality of the decomposition.

Because $\mathbf{R}$ is the analytic leftover of Figure 6.1 rather than a segmented highlight region, reducing α suppresses everything the diffuse product fails to explain, diffuse surfaces included, not one isolated highlight. Residual scaling is therefore a targeted edit on a predicted factor, not a semantic glare mask or a general-purpose glare remover; how selective it appears is governed entirely by the quality of the decomposition.

Illumination editing. Classic brightness or contrast adjustments operate on the entangled image, so lifting dark regions also washes out material texture and colour. With the decomposition, tonal edits can be applied to the shading layer alone:

$$\mathbf{I}'_f = \mathbf{A}_d \odot f(\mathbf{S}_d) + \mathbf{R}, \quad (6.2)$$

where f is a monotone luminance gain. Because our shading is unbounded, we do not apply $\mathbf{S}_d^\gamma$ directly. We normalise shading luminance by its 95th percentile q_{95} and use

$$f_\gamma(\mathbf{S}_d) = \mathbf{S}_d \left[\text{clip} \left(\frac{L(\mathbf{S}_d)}{q_{95}}, \varepsilon, 1 \right) \right]^{\gamma-1}, \quad 0 < \gamma < 1, \quad (6.3)$$

with the gain capped at 2.5. Low predicted illumination is lifted while values at or above the reference percentile are unchanged. Because $\mathbf{A}_d$ is untouched, the model’s predicted material colours and texture are held fixed while its predicted illumination changes. This is where the invariance targeted throughout the thesis pays off directly: an albedo that does not carry the illuminant’s colour can be re-lit without its materials drifting in hue, so lifting a dark, warm-lit corner reveals the surfaces’ own colour rather than amplifying the lamp’s cast. The result is a shading edit built directly on the decomposition, and its fidelity is bounded by how completely that cast was removed in the first place.

By construction this holds the predicted albedo of Figure 6.1 fixed, whereas the same luminance gain applied to the entangled image multiplies material and illumination together and so shifts material appearance as well. The edit is therefore a statement about controllability of the predicted factors, not about ground-truth preservation.

Both edits succeed exactly to the degree the decomposition does, and their two failure modes map onto the two axes measured in Chapter 5. Wherever some illuminant colour has leaked into the albedo (the residual hue drift measured in Sect. 5.7), that component is already baked into the material layer and will not

respond to a shading edit. And wherever the shading is structurally soft (a sharp shadow boundary that was never fully separated from the albedo), scaling the shading cannot cleanly remove it. Neither limitation is repaired by the editing operation; each is inherited from the factors being edited, which is why the constancy and structure results of the previous chapter are what ultimately bound how far this application can be taken.

7 | Conclusion

7.1 Summary

This report studied Cross-Illumination Albedo Invariance (CIAI), a training strategy for intrinsic image decomposition in which two photographs of the same scene under different illumination must yield the same albedo, and the shading must explain the change in brightness between them. We added it as a second training stage to a feed-forward model with a frozen DINOv2-Large encoder, a DPT decoder and three-channel shading, using real raw pairs from the Multi-Illumination Dataset and the MIDIntrinsics pseudo-ground truth.

The strategy does what its training signal supports. MID’s pairs differ mainly in the direction and intensity of a white flash, and the explanation loss is defined on luminance; correspondingly, CIAI reduces the variation of recovered material lightness across illuminations by 28% and 39% in matched ablations, and it lowers the dense albedo error on raw, coloured-lit ARAP renderings by 7–14%, a benchmark outside its training data. Against other methods our model is more lightness-stable than CD-IID, both Marigold variants and Ordinal Shading, level with CRefNet and behind RGB→X, while training 18.5M parameters. It is behind CD-IID on in-domain chroma error and mid-table on MAW.

The strategy does not deliver colour stability. Hue drift improves only slightly under CIAI, and our model’s drift grows with the illuminant colour as much as a grey-shading method’s, which is what a luminance-only explanation loss and MID’s white flash predict. The method that does show colour-independent drift, CD-IID, predicts the shading colour in a dedicated stage, which suggests a structural remedy: a model that separates shading luminance from shading chroma, planned below.

The colour path of the base model is a design choice that restores the albedo colour the frozen encoder tends to wash out, and it is separate from CIAI. Every stability result here is reported beside an accuracy result, because a grey, constant albedo is perfectly stable.

7.2 Limitations

Colour. The main limitation is that the trained model does not separate illuminant colour from material colour reliably. The explanation loss uses luminance only, and the real training pairs vary colour over a median gap of 0.030; the stability benchmark measures hue drift over the same narrow range, so the colour results are both a modelling result and a measurement limit.

One architecture, single runs. CIAI is presented as a training strategy, but it has been tested on one architecture, a frozen-encoder regression model. Whether its lightness gain holds for convolutional, fine-tuned transformer or diffusion models is untested. Most comparisons are single training runs, so small differences should not be read as effects.

Benchmarks. The ARAP results were computed with a mask that fails on ARAP’s mixed ground-truth encodings and are provisional (Sect. 5.5). MID is in the training data of both our model and CD-IID, so no MID comparison is zero-shot, and its pseudo-ground truth is itself derived from the capture stack.

Single-image accuracy. CIAI buys stability, not single-image accuracy: on MAW it slightly worsens both measures, and on IIW the model trails most baselines. A stronger base model, or the cross-architecture study below, is needed before the strategy can be judged on accuracy.

Achromatic ambiguity and non-diffuse surfaces. Hard cast shadows can still be read as material, and mirrors, glass and glossy plastic are unstable across illuminations because an albedo–shading product has no representation for them; the residual absorbs only part of their appearance. Figure 7.1 shows the mirror sphere of a MID scene: its reflections move with the flash in the input and, in the same way, in our albedo.

7.3 Future Work

The next stage of the project follows directly from the findings above. It has five parts, ordered so that each result is interpretable before the next is built on it.

1. Show that CIAI is a training strategy, not a property of one decoder. Train each of three architectures with and without the CIAI stage, from the same supervised initialisation, on the same data and for the same number of steps, with

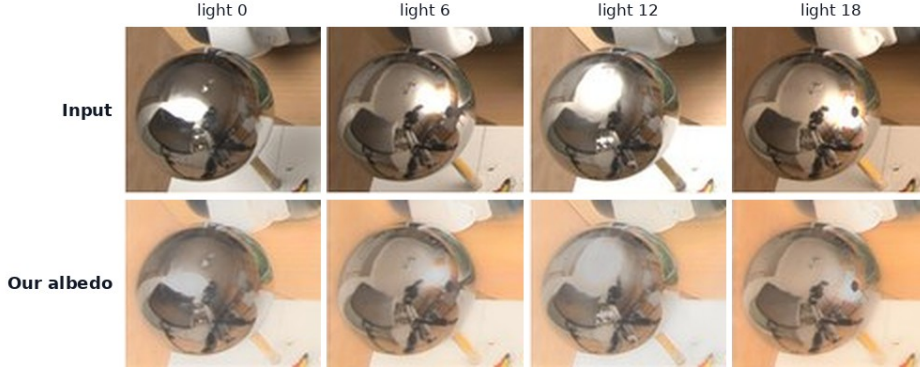


Figure 7.1: A mirror sphere under four flash directions (top) and our predicted albedo for each (bottom). The reflections change with the light in both rows, because a diffuse albedo–shading product cannot represent a mirror. Rendered with a later checkpoint of the same model (see Sect. 5.1).

three seeds per arm: a convolutional U-Net with an ImageNet-pretrained encoder; a DPT decoder on a *fine-tuned* DINOv2 encoder; and a latent-diffusion decomposer (Marigold-IID) adapted to single-step deterministic inference (Martin Garcia et al., 2025), which makes the prediction a differentiable function of the input so that a paired loss can be applied to it. The claim to test is a consistent reduction in C_{mat} without loss on MAW.

2. A shading model that separates luminance from chroma. Predict the image as

$$\mathbf{I} = \mathbf{A} \odot (S_{\text{lum}} \mathbf{C}) + \mathbf{R}, \quad Y(\mathbf{C}) = 1, \quad (7.1)$$

with a scalar shading luminance S_{lum} , a unit-luminance shading chroma map $\mathbf{C}$ predicted by its own head with a low-frequency and a global component, and the albedo $\mathbf{A}$. This follows the separation that makes CD-IID colour-stable, inside a single network (`src/models/v21_trifactor.py`). The colour path changes accordingly: the albedo head receives the illuminant-corrected image $\mathbf{I}/(S_{\text{lum}} \mathbf{C})$ rather than the raw input, so that material colour reaches it and illuminant colour does not. A three-row ablation (raw skip, corrected skip, no skip, each with the chroma loss on albedo) decides whether the corrected skip is better.

3. Losses matched to the factors. Keep $\mathcal{L}_{\text{inv}}$ on the albedo and apply the luminance explanation loss to S_{lum} . Add a chroma explanation loss on $\mathbf{C}$. With $\mathbf{r} = \log(\mathbf{x}_1 + \varepsilon) - \log(\mathbf{x}_2 + \varepsilon)$ per channel and $c(\mathbf{r}) = \mathbf{r} - \frac{1}{3} \sum_c r_c$ its chromatic part, it asks $c(\mathbf{r}_{\mathbf{C}})$ to match $c(\mathbf{r}_{\mathbf{I}})$ over the valid pixels, which scores exactly the component that Eq. (3.9) discards and can only be satisfied by moving colour

into the chroma factor. Apply it only to pairs whose measured illuminant gap exceeds a threshold (for example 0.08), since on near-neutral pairs it would fit noise. Supervise $\mathbf{C}$ directly wherever ground truth exists.

4. Training pairs whose illuminant colour actually varies. A rendered corpus built from 3D-FRONT covers the range from near-neutral to strongly coloured pairs (tenth to ninetieth percentile gap 0.008 to 0.35) with exact albedo and shading chroma; its pilot fails only a hard-shadow quality gate, which should not block its use for colour. Real coloured pairs are the other half: the LSMI dataset (Kim et al., 2021) photographs each scene under several combinations of two or three illuminants and provides per-pixel illuminant chromaticity, which yields both cross-illumination pairs and a direct target for $\mathbf{C}$. If neither transfers, a small capture of our own with coloured filters on an MID-style rig would provide pixel-aligned real pairs with known colour changes.

5. Evaluation on a real colour range. Regenerate ARAP with encoding-aware masks; report hue drift on the ARAP pairs whose illuminant colour changes by at least 0.15 (26 pairs in 17 scenes), on held-out LSMI scenes and on held-out 3D-Front pairs, always beside MAW; and report the absolute drift together with the relative one, so that a reduction in drift accompanied by a loss of colour is visible in both.

Applications. Chapter 6 demonstrated qualitative factor-space edits. Once the shading chroma is a separate factor, relighting and white-balance edits become direct operations on $\mathbf{C}$, and evaluating them quantitatively, for example by user study, becomes worthwhile.

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List of Figures

1.1	Visual meaning of an intrinsic decomposition.	2
1.2	MID albedo predictions under four flash directions.	3
2.1	Why intrinsic decomposition is ambiguous.	8
3.1	Why scalar shading leaks illuminant colour into albedo.	16
3.2	Inverse-shading parameterisation and target distributions.	18
3.3	Model architecture and supervision.	21
3.4	Two supervision lanes: a white-balanced anchor and a raw CIAI pair.	24
3.5	Cross-Illumination Albedo Invariance during training.	24
4.1	$\mathcal{L}_a$ versus $\mathcal{L}_{\text{chroma}}$: what each anchor actually penalises.	27
4.2	One representative frame per dataset, with its training role.	30
4.3	MID training preprocessing paths.	31
4.4	Two-stage training schedule.	32
5.1	Inference-resolution comparison on MAW.	39
5.2	The four evaluation benchmarks and their supervision.	40
5.3	Qualitative IIW albedo comparison across methods.	41
5.4	C_{mat} : four illuminants, one material.	42
5.5	Cast_{rel} : one material’s chroma across illuminants.	43
5.6	$\text{Chroma}_{\text{err}}$: stable is not the same as correct.	44
5.7	Spread ratio: materials keep different colours.	45
5.8	MID hue stability versus pseudo-GT chroma error.	47
5.9	MID chroma calibration and spread-collapse diagnostic.	48
5.10	MID albedo stability under four flash directions.	48
5.11	IIW qualitative predictions and failure cases.	51
5.12	Qualitative MAW albedo comparison.	54
5.13	ARAP model comparison across illuminants on <i>camera</i>	56
5.14	ARAP qualitative accuracy on three indoor scenes.	59
5.15	Qualitative CIAI ablation on MID.	63

LIST OF FIGURES

6.1	Predicted decomposition of a realistic indoor photograph.	68
7.1	Reflective surfaces stay unstable.	73

List of Tables

4.1	Training loss terms and weights.	26
5.1	Methods, model size, and inference cost.	38
5.2	MID stability and chroma accuracy.	46
5.3	Paired comparisons against each external baseline.	46
5.4	IIW WHDR results.	50
5.5	MAW albedo accuracy results.	53
5.6	ARAP cross-illumination constancy (provisional).	57
5.7	ARAP dense-albedo accuracy (provisional).	58
5.8	CIAI ablation across constancy and accuracy.	61
5.9	Illuminant colour change between paired frames.	64
5.10	Does hue drift follow illuminant colour?	64